I_{DDS}/V_P

CHAPTER

Field-Effect Transistors 5

5.1 INTRODUCTION

The field-effect transistor (FET) is a three-terminal device used for a variety of applications that match, to a large extent, those of the BJT transistor described in Chapters 3 and 4. Although there are important differences between the two types of devices, there are also many similarities that will be pointed out in the sections to follow.

The primary difference between the two types of transistors is the fact that the BJT transistor is a *current-controlled* device as depicted in Fig. 5.1a, while the JFET transistor is a *voltage-controlled* device as shown in Fig. 5.1b. In other words, the current I_C in Fig. 5.1a is a direct function of the level of I_B. For the FET the current I will be a function of the voltage V_{GS} applied to the input circuit as shown in Fig. 5.1b. In each case the current of the output circuit is being controlled by a parameter of the input circuit—in one case a current level and in the other an applied voltage.

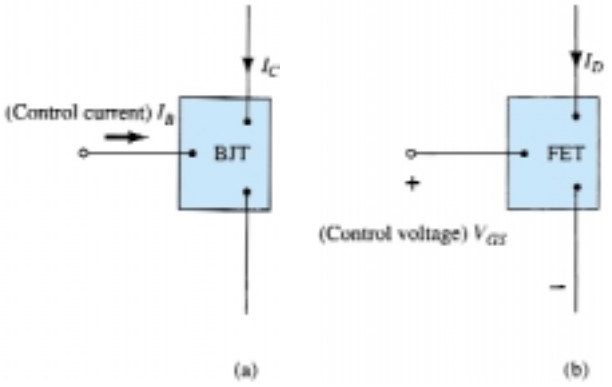


Figure 5.1 (a) Current-controlled and (b) voltage-controlled amplifiers.

Just as there are *npn* and *pnp* bipolar transistors, there are *n-channel* and *p-channel* field-effect transistors. However, it is important to keep in mind that the BJT transistor is a *bipolar* device—the prefix *bi-* revealing that the conduction level is a function of two charge carriers, electrons and holes. The FET is a *unipolar* device depending solely on either electron (*n*-channel) or hole (*p*-channel) conduction.

The term field-effect in the chosen name deserves some explanation. We are all familiar with the ability of a permanent magnet to draw metal filings to the magnet without the need for actual contact. The magnetic field of the permanent magnet has enveloped the filings and attracted them to the magnet through an effort on the part of the magnetic flux lines to be as short as possible. For the FET an *electric field* is established by the charges present that will control the conduction path of the output

circuit without the need for direct contact between the controlling and controlled quantities.

There is a natural tendency when introducing a second device with a range of applications similar to one already introduced to compare some of the general characteristics of one versus the other. One of the most important characteristics of the FET is its *high input impedance.* At a level of 1 to several hundred megohms it far exceeds the typical input resistance levels of the BJT transistor configurations—a very important characteristic in the design of linear ac amplifier systems. On the other hand, the BJT transistor has a much higher sensitivity to changes in the applied signal. In other words, the variation in output current is typically a great deal more for BJTs than FETs for the same change in applied voltage. For this reason, typical ac voltage gains for BJT amplifiers are a great deal more than for FETs. In general, FETs are more temperature stable than BJTs, and FETs are usually smaller in construction than BJTs, making them particularly useful in *integrated-circuit (IC)* chips. The construction characteristics of some FETs, however, can make them more sensitive to handling than BJTs.

Two types of FETs will be introduced in this chapter: the *junction field-effect transistor* (JFET) and the *metal-oxide-semiconductor field-effect transistor (MOSFET).* The MOSFET category is further broken down into depletion and enhancement types, which are both described. The MOSFET transistor has become one of the most important devices used in the design and construction of integrated circuits for digital computers. Its thermal stability and other general characteristics make it extremely popular in computer circuit design. However, as a discrete element in a typical top-hat container, it must be handled with care (to be discussed in a later section).

Once the FET construction and characteristics have been introduced, the biasing arrangements will be covered in Chapter 6. The analysis performed in Chapter 4 using BJT transistors will prove helpful in the derivation of the important equations and understanding the results obtained for FET circuits.



Drs. Ian Munro Ross (front) and G. C. Dacey jointly developed an experimental procedure for measuring the characteristics of a field-effect transistor in 1955. (Courtesy of AT&T Archives.)

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PhD Gonville and Caius College, Cambridge
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Chairman—National Advisory Committee on Semiconductors

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Vice President, Research at Sandia Corporation
Member IRE, Tau Beta Pi, Eta Kappa Nu

5.2 CONSTRUCTION AND CHARACTERISTICS OF JFETs

As indicated earlier, the JFET is a three-terminal device with one terminal capable of controlling the current between the other two. In our discussion of the BJT transistor the *npn* transistor was employed through the major part of the analysis and design sections, with a section devoted to the impact of using a *pnp* transistor. For the JFET transistor the *n*-channel device will appear as the prominent device, with paragraphs and sections devoted to the impact of using a *p*-channel JFET.

The basic construction of the *n*-channel JFET is shown in Fig. 5.2. Note that the major part of the structure is the *n*-type material that forms the channel between the embedded layers of *p*-type material. The top of the *n*-type channel is connected through an ohmic contact to a terminal referred to as the *drain (D),* while the lower end of the same material is connected through an ohmic contact to a terminal referred to as the *source (S).* The two *p*-type materials are connected together and to the *gate (G)* terminal. In essence, therefore, the drain and source are connected to the ends of the *n*-type channel and the gate to the two layers of *p*-type material. In the absence of any applied potentials the JFET has two *p-n* junctions under no-bias conditions. The result is a depletion region at each junction as shown in Fig. 5.2 that resembles the same region of a diode under no-bias conditions. Recall also that a depletion region is that region void of free carriers and therefore unable to support conduction through the region.

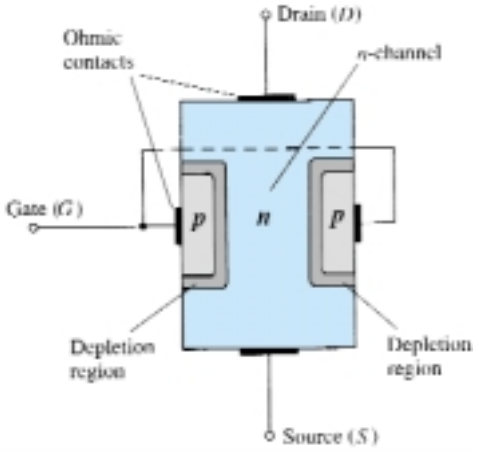


Figure 5.2 Junction field-effect transistor (JFET).

Analogies are seldom perfect and at times can be misleading, but the water analogy of Fig. 5.3 does provide a sense for the JFET control at the gate terminal and the appropriateness of the terminology applied to the terminals of the device. The source of water pressure can be likened to the applied voltage from drain to source that will establish a flow of water (electrons) from the spigot (source). The "gate," through an applied signal (potential), controls the flow of water (charge) to the "drain." The drain and source terminals are at opposite ends of the *n*-channel as introduced in Fig. 5.2 because the terminology is defined for electron flow.



Figure 5.3 Water analogy for the JFET control mechanism.

$V_{GS} = 0$ V, V_{DS} Some Positive Value

In Fig. 5.4, a positive voltage V_{DS} has been applied across the channel and the gate has been connected directly to the source to establish the condition $V_{GS} = 0$ V. The result is a gate and source terminal at the same potential and a depletion region in the low end of each *p*-material similar to the distribution of the no-bias conditions of Fig. 5.2. The instant the voltage V_{DD} ($= V_{DS}$) is applied, the electrons will be drawn to the drain terminal, establishing the conventional current I_D with the defined direction of Fig. 5.4. The path of charge flow clearly reveals that the drain and source currents are equivalent ($I_D = I_S$). Under the conditions appearing in Fig. 5.4, the flow of charge is relatively uninhibited and limited solely by the resistance of the *n*-channel between drain and source.

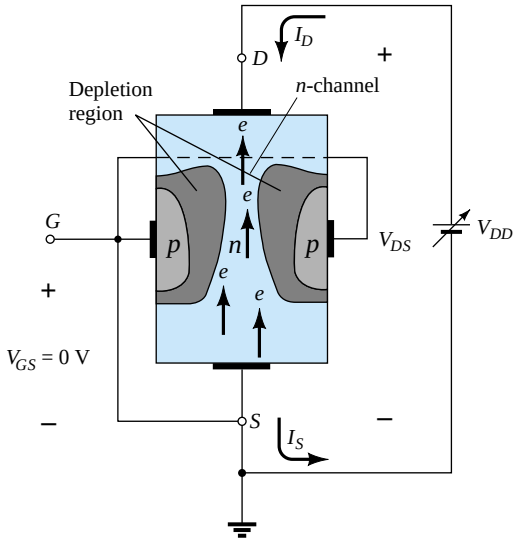


Figure 5.4 JFET in the $V_{GS} = 0$ V and $V_{DS} > 0$ V.

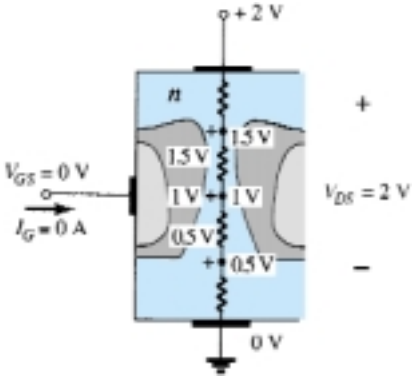


Figure 5.5 Varying reverse-bias potentials across the *p-n* junction of an *n*-channel JFET.

It is important to note that the depletion region is wider near the top of both *p*-type materials. The reason for the change in width of the region is best described through the help of Fig. 5.5. Assuming a uniform resistance in the *n*-channel, the resistance of the channel can be broken down to the divisions appearing in Fig. 5.5. The current I_D will establish the voltage levels through the channel as indicated on the same figure. The result is that the upper region of the *p*-type material will be reverse-biased by about 1.5 V, with the lower region only reverse-biased by 0.5 V. Recall from the discussion of the diode operation that the greater the applied reverse bias, the wider the depletion region—hence the distribution of the depletion region as shown in Fig. 5.5. The fact that the *p-n* junction is reverse-biased for the length of the channel results in a gate current of zero amperes as shown in the same figure. The fact that $I_G = 0$ A is an important characteristic of the JFET.

As the voltage V_{DS} is increased from 0 to a few volts, the current will increase as determined by Ohm's law and the plot of I_D versus V_{DS} will appear as shown in Fig. 5.6. The relative straightness of the plot reveals that for the region of low values of V_{DS}, the resistance is essentially constant. As V_{DS} increases and approaches a level referred to as V_P in Fig. 5.6, the depletion regions of Fig. 5.4 will widen, causing a noticeable reduction in the channel width. The reduced path of conduction causes the resistance to increase and the curve in the graph of Fig. 5.6 to occur. The more horizontal the curve, the higher the resistance, suggesting that the resistance is approaching "infinite" ohms in the horizontal region. If V_{DS} is increased to a level where it appears that the two depletion regions would "touch" as shown in Fig. 5.7, a condition referred to as *pinch-off* will result. The level of V_{DS} that establishes this condition is referred to as the *pinch-off voltage* and is denoted by V_P as shown in Fig. 5.6. In actuality, the term *pinch-off* is a misnomer in that it suggests the current I_D is pinched off and drops to 0 A. As shown in Fig. 5.6, however, this is hardly the case — I_D maintains a saturation level defined as I_{DSS} in Fig. 5.6. In reality a very small channel still exists, with a current of very high density. The fact that I_D does not drop off at pinch-off and maintains the saturation level indicated in Fig. 5.6 is verified by the following fact: The absence of a drain current would remove the possibility of different potential levels through the *n*-channel material to establish the varying levels of reverse bias along the *p-n* junction. The result would be a loss of the depletion region distribution that caused pinch-off in the first place.

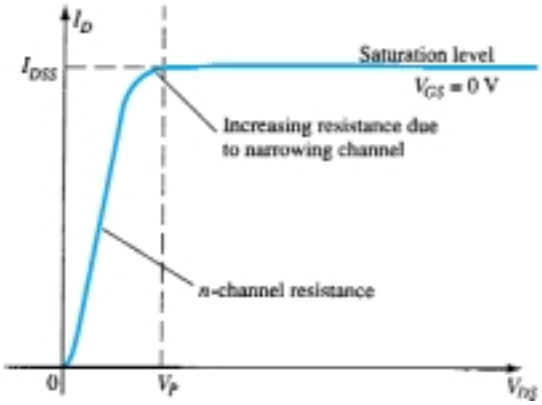


Figure 5.6 I_D versus V_{DS} for $V_{GS} = 0$ V.

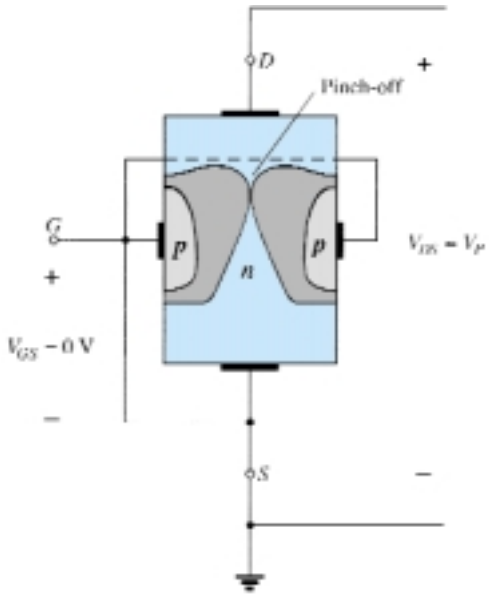


Figure 5.7 Pinch-off ($V_{GS} = 0$ V, $V_{DS} = V_P$).

As V_{DS} is increased beyond V_P, the region of close encounter between the two depletion regions will increase in length along the channel, but the level of I_D remains essentially the same. In essence, therefore, once $V_{DS} > V_P$ the JFET has the characteristics of a current source. As shown in Fig. 5.8, the current is fixed at $I_D = I_{DSS}$, but the voltage V_{DS} (for levels $> V_P$) is determined by the applied load.

The choice of notation I_{DSS} is derived from the fact that it is the *D*rain-to-*S*ource current with a *S*hort-circuit connection from gate to source. As we continue to investigate the characteristics of the device we will find that:

I_{DSS} is the maximum drain current for a JFET and is defined by the conditions $V_{GS} = 0$ V and $V_{DS} > |V_P|$.

Note in Fig. 5.6 that $V_{GS} = 0$ V for the entire length of the curve. The next few paragraphs will describe how the characteristics of Fig. 5.6 are affected by changes in the level of V_{GS}.

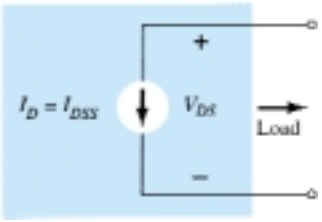


Figure 5.8 Current source equivalent for $V_{GS} = 0$ V, $V_{DS} > V_P$.

$V_{GS} < 0$ V

The voltage from gate to source, denoted V_{GS}, is the controlling voltage of the JFET. Just as various curves for I_C versus V_{CE} were established for different levels of I_B for the BJT transistor, curves of I_D versus V_{DS} for various levels of V_{GS} can be developed for the JFET. For the *n*-channel device the controlling voltage V_{GS} is made more and more negative from its $V_{GS} = 0$ V level. In other words, the gate terminal will be set at lower and lower potential levels as compared to the source.

In Fig. 5.9 a negative voltage of -1 V has been applied between the gate and source terminals for a low level of V_{DS}. The effect of the applied negative-bias V_{GS} is to establish depletion regions similar to those obtained with $V_{GS} = 0$ V but at lower levels of V_{DS}. Therefore, the result of applying a negative bias to the gate is to reach the saturation level at a lower level of V_{DS} as shown in Fig. 5.10 for $V_{GS} = -1$ V. The resulting saturation level for I_D has been reduced and in fact will continue to decrease as V_{GS} is made more and more negative. Note also on Fig. 5.10 how the pinch-off voltage continues to drop in a parabolic manner as V_{GS} becomes more and more negative. Eventually, V_{GS} when $V_{GS} = -V_P$ will be sufficiently negative to establish a saturation level that is essentially 0 mA, and for all practical purposes the device has been "turned off." In summary:

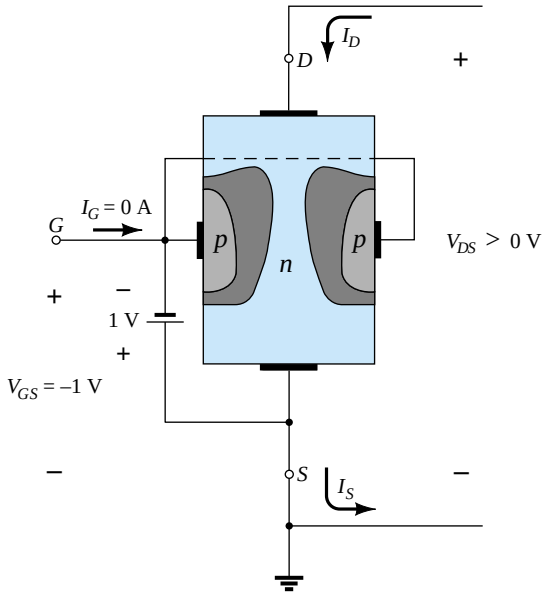


Figure 5.9 Application of a negative voltage to the gate of a JFET.

I_{DDS}/V_P

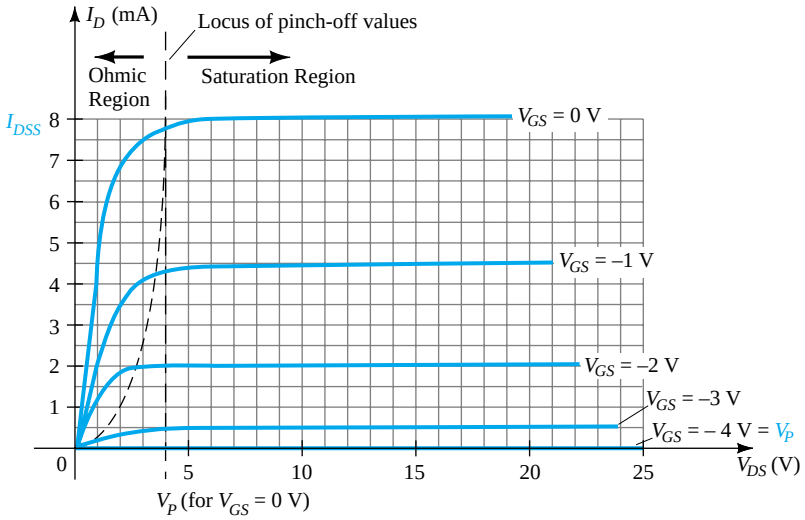


Figure 5.10 *n*-Channel JFET characteristics with $I_{DSS} = 8$ mA and $V_P = -4$ V.

The level of V_{GS} that results in $I_D = 0$ mA is defined by $V_{GS} = V_P$, with V_P being a negative voltage for n-channel devices and a positive voltage for p-channel JFETs.

On most specification sheets the pinch-off voltage is specified as $V_{GS(off)}$ rather than V_P. A specification sheet will be reviewed later in the chapter when the primary elements of concern have been introduced. The region to the right of the pinch-off locus of Fig. 5.10 is the region typically employed in linear amplifiers (amplifiers with minimum distortion of the applied signal) and is commonly referred to as the *constant-current, saturation,* or *linear amplification region.*

Voltage-Controlled Resistor

The region to the left of the pinch-off locus of Fig. 5.10 is referred to as the *ohmic* or *voltage-controlled resistance region.* In this region the JFET can actually be employed as a variable resistor (possibly for an automatic gain control system) whose resistance is controlled by the applied gate-to-source voltage. Note in Fig. 5.10 that the slope of each curve and therefore the resistance of the device between drain and source for $V_{DS} < V_P$ is a function of the applied voltage V_{GS}. As V_{GS} becomes more and more negative, the slope of each curve becomes more and more horizontal, corresponding with an increasing resistance level. The following equation will provide a good first approximation to the resistance level in terms of the applied voltage V_{GS}.

$$r_d = \frac{r_o}{(1 - V_{GS}/V_P)^2} \tag{5.1}$$

where r_o is the resistance with $V_{GS} = 0$ V and r_d the resistance at a particular level of V_{GS}.

For an *n*-channel JFET with r_o equal to 10 kΩ ($V_{GS} = 0$ V, $V_P = -6$ V), Eq. (5.1) will result in 40 kΩ at $V_{GS} = -3$ V.

p-Channel Devices

The *p*-channel JFET is constructed in exactly the same manner as the *n*-channel device of Fig. 5.2, but with a reversal of the *p*- and *n*-type materials as shown in Fig. 5.11.

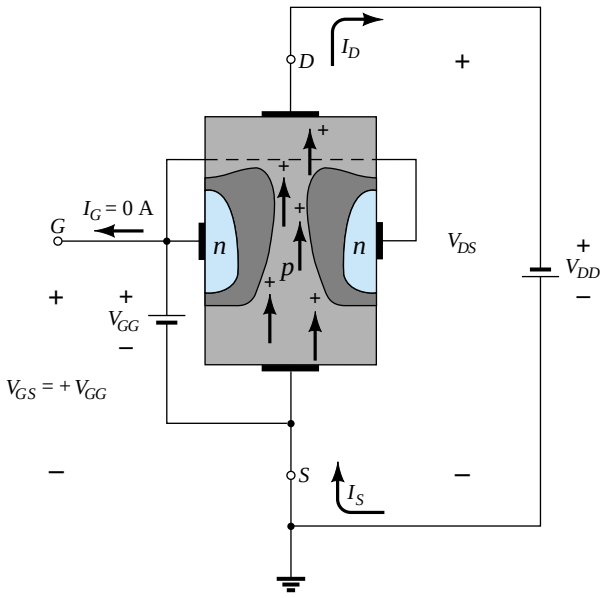


Figure 5.11 *p*-Channel JFET.

The defined current directions are reversed, as are the actual polarities for the voltages V_{GS} and V_{DS}. For the *p*-channel device, the channel will be constricted by increasing positive voltages from gate to source and the double-subscript notation for V_{DS} will result in negative voltages for V_{DS} on the characteristics of Fig. 5.12, which has an I_{DSS} of 6 mA and a pinch-off voltage of $V_{GS} = +6$ V. Do not let the minus signs for V_{DS} confuse you. They simply indicate that the source is at a higher potential than the drain.

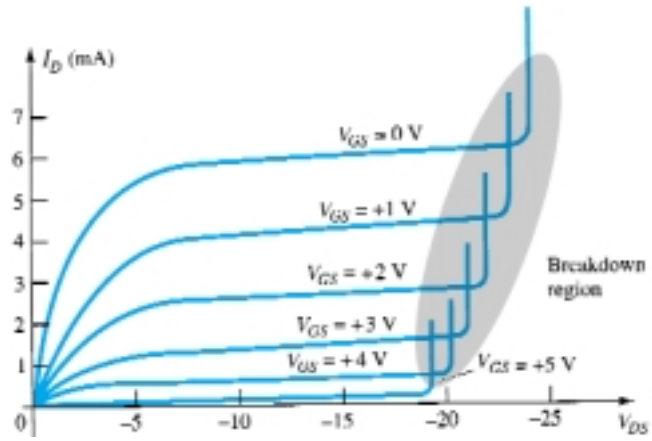


Figure 5.12 *p*-Channel JFET characteristics with $I_{DSS} = 6$ mA and $V_P = +6$ V.

Note at high levels of V_{DS} that the curves suddenly rise to levels that seem unbounded. The vertical rise is an indication that breakdown has occurred and the current through the channel (in the same direction as normally encountered) is now limited solely by the external circuit. Although not appearing in Fig. 5.10 for the *n*-channel device, they do occur for the *n*-channel device if sufficient voltage is applied. This region can be avoided if the level of $V_{DS_{max}}$ is noted on the specification sheet and the design is such that the actual level of V_{DS} is less than this value for *all* values of V_{GS}.

Symbols

The graphic symbols for the *n*-channel and *p*-channel JFETs are provided in Fig. 5.13. Note that the arrow is pointing in for the *n*-channel device of Fig. 5.13a to represent the direction in which I_G would flow if the *p*-*n* junction were forward-biased. For the *p*-channel device (Fig. 5.13b) the only difference in the symbol is the direction of the arrow.

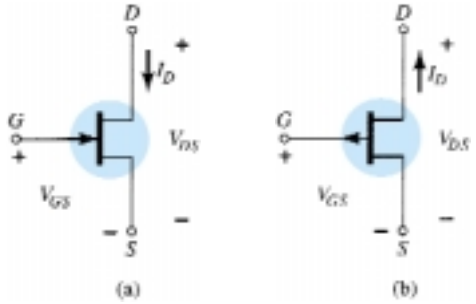


Figure 5.13 JFET symbols: (a) *n*-channel; (b) *p*-channel.

Summary

A number of important parameters and relationships were introduced in this section. A few that will surface frequently in the analysis to follow in this chapter and the next for *n*-channel JFETs include the following:

The maximum current is defined as I_{DSS} and occurs when $V_{GS} = 0$ V and $V_{DS} \geq |V_P|$ as shown in Fig. 5.14a.

For gate-to-source voltages V_{GS} less than (more negative than) the pinch-off level, the drain current is 0 A ($I_D = 0$ A) as appearing in Fig. 5.14b.

For all levels of V_{GS} between 0 V and the pinch-off level, the current I_D will range between I_{DSS} and 0 A, respectively, as reviewed by Fig. 5.14c.

For p-channel JFETs a similar list can be developed.

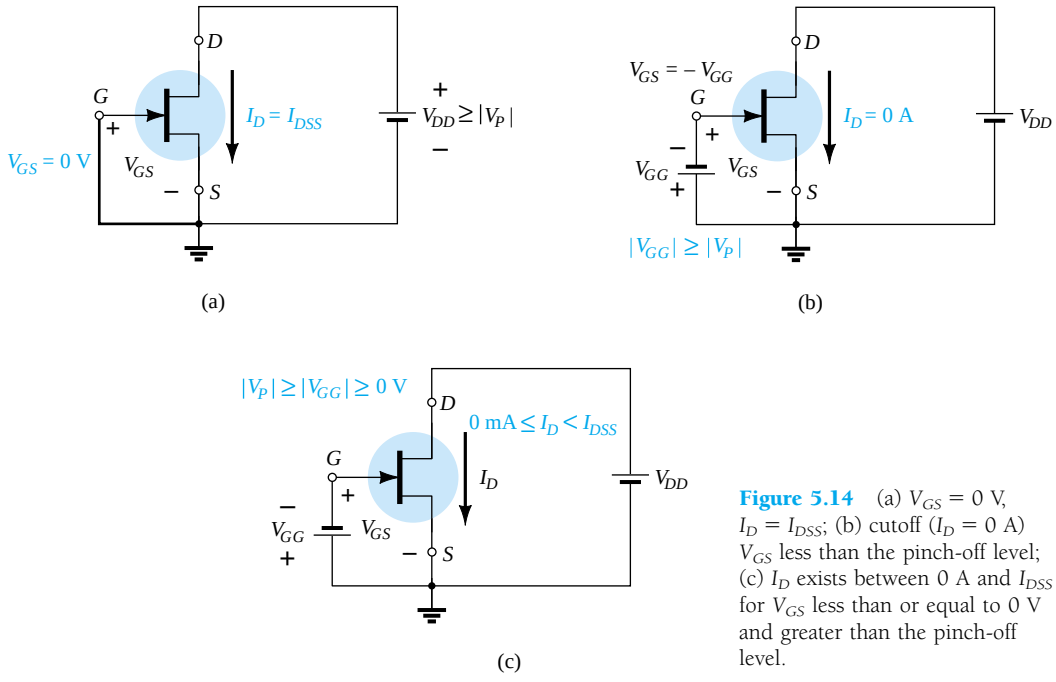


Figure 5.14 (a) $V_{GS} = 0$ V, $I_D = I_{DSS}$; (b) cutoff ($I_D = 0$ A) V_{GS} less than the pinch-off level; (c) I_D exists between 0 A and I_{DSS} for V_{GS} less than or equal to 0 V and greater than the pinch-off level.

5.3 TRANSFER CHARACTERISTICS

Derivation

For the BJT transistor the output current I_C and input controlling current I_B were related by beta, which was considered constant for the analysis to be performed. In equation form,

$$I_C = f(I_B) = \beta I_B \qquad (5.2)$$

(control variable; constant)

In Eq. (5.2) a linear relationship exists between I_C and I_B. Double the level of I_B and I_C will increase by a factor of two also.

Unfortunately, this linear relationship does not exist between the output and input quantities of a JFET. The relationship between I_D and V_{GS} is defined by *Shockley's equation:*

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2 \qquad (5.3)$$

(control variable; constants)

The squared term of the equation will result in a nonlinear relationship between I_D and V_{GS}, producing a curve that grows exponentially with decreasing magnitudes of V_{GS}.

For the dc analysis to be performed in Chapter 6, a graphical rather than mathematical approach will in general be more direct and easier to apply. The graphical approach, however, will require a plot of Eq. (5.3) to represent the device and a plot of the network equation relating the same variables. The solution is defined by the point of intersection of the two curves. It is important to keep in mind when applying the graphical approach that the device characteristics will be *unaffected* by the network in which the device is employed. The network equation may change along with the intersection between the two curves, but the transfer curve defined by Eq. (5.3) is unaffected. In general, therefore:

The transfer characteristics defined by Shockley's equation are unaffected by the network in which the device is employed.

The transfer curve can be obtained using Shockley's equation or from the output characteristics of Fig. 5.10. In Fig. 5.15 two graphs are provided, with the vertical

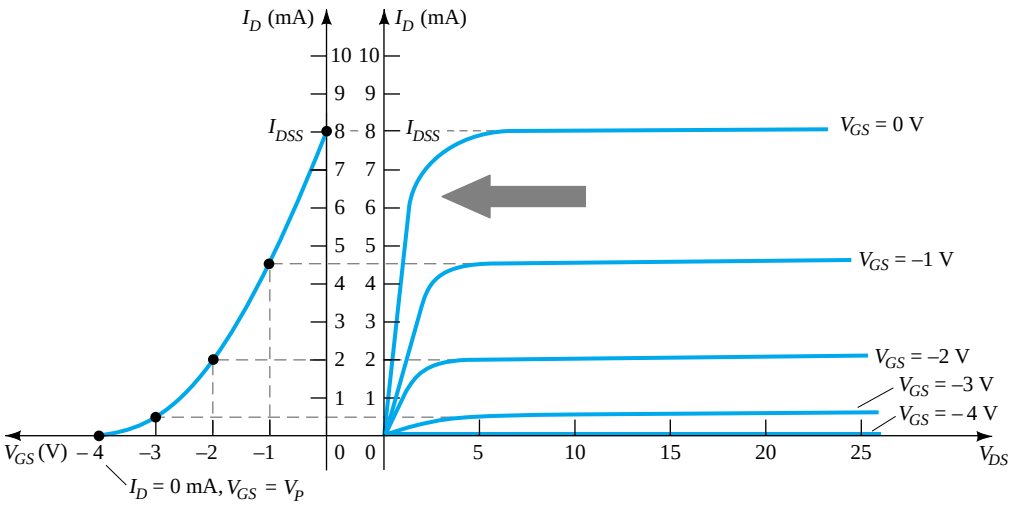
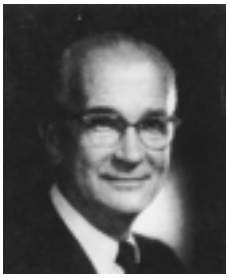


Figure 5.15 Obtaining the transfer curve from the drain characteristics.



William Bradford Shockley (1910–1989), co-inventor of the first transistor and formulator of the "field-effect" theory employed in the development of the transistor and FET. (Courtesy of AT&T Archives.)

Born: London, England
PhD Harvard, 1936
Head, Transistor Physics Department–Bell Laboratories
President, Shockley Transistor Corp.
Poniatoff Professor of Engineering Science at Stanford University
Nobel Prize in physics in 1956 with Drs. Brattain and Bardeen

scaling in milliamperes for each graph. One is a plot of I_D versus V_{DS}, while the other is I_D versus V_{GS}. Using the drain characteristics on the right of the "y" axis, a horizontal line can be drawn from the saturation region of the curve denoted $V_{GS} = 0$ V to the I_D axis. The resulting current level for both graphs is I_{DSS}. The point of intersection on the I_D versus V_{GS} curve will be as shown since the vertical axis is defined as $V_{GS} = 0$ V.

In review:

When $V_{GS} = 0$ V, $I_D = I_{DSS}$.

When $V_{GS} = V_P = -4$ V, the drain current is zero milliamperes, defining another point on the transfer curve. That is:

When $V_{GS} = V_P$, $I_D = 0$ mA.

Before continuing, it is important to realize that the drain characteristics relate one output (or drain) quantity to another output (or drain) quantity—both axes are defined by variables in the same region of the device characteristics. The transfer characteristics are a plot of an output (or drain) current versus an input-controlling quantity. There is therefore a direct "transfer" from input to output variables when employing the curve to the left of Fig. 5.15. If the relationship were linear, the plot of I_D versus V_{GS} would result in a straight line between I_{DSS} and V_P. However, a parabolic curve will result because the vertical spacing between steps of V_{GS} on the drain characteristics of Fig. 5.15 decreases noticeably as V_{GS} becomes more and more negative. Compare the spacing between $V_{GS} = 0$ V and $V_{GS} = -1$ V to that between $V_{GS} = -3$ V and pinch-off. The change in V_{GS} is the same, but the resulting change in I_D is quite different.

If a horizontal line is drawn from the $V_{GS} = -1$ V curve to the I_D axis and then extended to the other axis, another point on the transfer curve can be located. Note that $V_{GS} = -1$ V on the bottom axis of the transfer curve with $I_D = 4.5$ mA. Note in the definition of I_D at $V_{GS} = 0$ V and -1 V that the saturation levels of I_D are employed and the ohmic region ignored. Continuing with $V_{GS} = -2$ V and -3 V, the transfer curve can be completed. It is the transfer curve of I_D versus V_{GS} that will receive extended use in the analysis of Chapter 6 and not the drain characteristics of Fig. 5.15. The next few paragraphs will introduce a quick, efficient method of plotting I_D versus V_{GS} given only the levels of I_{DSS} and V_P and Shockley's equation.

Applying Shockley's Equation

The transfer curve of Fig. 5.15 can also be obtained directly from Shockley's equation (5.3) given simply the values of I_{DSS} and V_P. The levels of I_{DSS} and V_P define the limits of the curve on both axes and leave only the necessity of finding a few intermediate plot points. The validity of Eq. (5.3) as a source of the transfer curve of Fig. 5.15 is best demonstrated by examining a few specific levels of one variable and finding the resulting level of the other as follows:

Substituting $V_{GS} = 0$ V gives

$$\text{Eq. (5.3):} \quad I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

$$= I_{DSS}\left(1 - \frac{0}{V_P}\right)^2 = I_{DSS}(1 - 0)^2$$

and

$$\boxed{I_D = I_{DSS}\,\big|_{V_{GS} = 0\text{ V}}} \tag{5.4}$$

Substituting $V_{GS} = V_P$ yields

$$I_D = I_{DSS}\left(1 - \frac{V_P}{V_P}\right)^2$$
$$= I_{DSS}(1 - 1)^2 = I_{DSS}(0)$$
$$\boxed{I_D = 0\ \text{A}\big|_{V_{GS} = V_P}} \tag{5.5}$$

For the drain characteristics of Fig. 5.15, if we substitute $V_{GS} = -1$ V,

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$
$$= 8\ \text{mA}\left(1 - \frac{-1\ \text{V}}{-4\ \text{V}}\right)^2 = 8\ \text{mA}\left(1 - \frac{1}{4}\right)^2 = 8\ \text{mA}(0.75)^2$$
$$= 8\ \text{mA}(0.5625)$$
$$\mathbf{= 4.5\ mA}$$

as shown in Fig. 5.15. Note the care taken with the negative signs for V_{GS} and V_P in the calculations above. The loss of one sign would result in a totally erroneous result.

It should be obvious from the above that given I_{DSS} and V_P (as is normally provided on specification sheets) the level of I_D can be found for any level of V_{GS}. Conversely, by using basic algebra we can obtain [from Eq. (5.3)] an equation for the resulting level of V_{GS} for a given level of I_D. The derivation is quite straight forward and will result in

$$\boxed{V_{GS} = V_P\left(1 - \sqrt{\frac{I_D}{I_{DSS}}}\right)} \tag{5.6}$$

Let us test Eq. (5.6) by finding the level of V_{GS} that will result in a drain current of 4.5 mA for the device with the characteristics of Fig. 5.15.

$$V_{GS} = -4\ \text{V}\left(1 - \sqrt{\frac{4.5\ \text{mA}}{8\ \text{mA}}}\right)$$
$$= -4\ \text{V}(1 - \sqrt{0.5625}) = -4\ \text{V}(1 - 0.75)$$
$$= -4\ \text{V}(0.25)$$
$$\mathbf{= -1\ V}$$

as substituted in the above calculation and verified by Fig. 5.15.

Shorthand Method

Since the transfer curve must be plotted so frequently, it would be quite advantageous to have a shorthand method for plotting the curve in the quickest, most efficient manner while maintaining an acceptable degree of accuracy. The format of Eq. (5.3) is such that specific levels of V_{GS} will result in levels of I_D that can be memorized to provide the plot points needed to sketch the transfer curve. If we specify V_{GS} to be one-half the pinch-off value V_P, the resulting level of I_D will be the following, as determined by Shockley's equation:

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$
$$= I_{DSS}\left(\frac{1 - V_P/2}{V_P}\right)^2 = I_{DSS}\left(1 - \frac{1}{2}\right)^2 = I_{DSS}(0.5)^2$$
$$= I_{DSS}(0.25)$$

I_{DDS}/V_P

and $$I_D = \frac{I_{DSS}}{4}\bigg|_{V_{GS} = V_P/2} \tag{5.7}$$

Now it is important to realize that Eq. (5.7) is not for a particular level of V_P. It is a general equation for any level of V_P as long as $V_{GS} = V_P/2$. The result specifies that the drain current will always be one-fourth of the saturation level I_{DSS} as long as the gate-to-source voltage is one-half the pinch-off value. Note the level of I_D for $V_{GS} = V_P/2 = -4\text{ V}/2 = -2\text{ V}$ in Fig. 5.15.

If we choose $I_D = I_{DSS}/2$ and substitute into Eq. (5.6), we find that

$$V_{GS} = V_P\left(1 - \sqrt{\frac{I_D}{I_{DSS}}}\right)$$

$$= V_P\left(1 - \sqrt{\frac{I_{DSS}/2}{I_{DSS}}}\right) = V_P(1 - \sqrt{0.5}) = V_P(0.293)$$

and $$V_{GS} \cong 0.3V_P\big|_{I_D = I_{DSS}/2} \tag{5.8}$$

Additional points can be determined, but the transfer curve can be sketched to a satisfactory level of accuracy simply using the four plot points defined above and reviewed in Table 5.1. In fact, in the analysis of Chapter 6, a maximum of four plot points are used to sketch the transfer curves. On most occasions using just the plot point defined by $V_{GS} = V_P/2$ and the axis intersections at I_{DSS} and V_P will provide a curve accurate enough for most calculations.

TABLE 5.1 V_{GS} versus I_D Using Shockley's Equation

V_{GS}	I_D
0	I_{DSS}
0.3 V_P	$I_{DSS}/2$
0.5 V_P	$I_{DSS}/4$
V_P	0 *mA*

EXAMPLE 5.1 Sketch the transfer curve defined by $I_{DSS} = 12\text{ mA}$ and $V_P = -6\text{ V}$.

Solution

Two plot points are defined by

$$I_{DSS} = \mathbf{12\text{ mA}} \quad \text{and} \quad V_{GS} = \mathbf{0\text{ V}}$$

and $$I_D = \mathbf{0\text{ mA}} \quad \text{and} \quad V_{GS} = \boldsymbol{V_P}$$

At $V_{GS} = V_P/2 = -6\text{ V}/2 = \mathbf{-3\text{ V}}$ the drain current will be determined by $I_D = I_{DSS}/4 = 12\text{ mA}/4 = \mathbf{3\text{ mA}}$. At $I_D = I_{DSS}/2 = 12\text{ mA}/2 = \mathbf{6\text{ mA}}$ the gate-to-source voltage is determined by $V_{GS} \cong 0.3V_P = 0.3(-6\text{ V}) = \mathbf{-1.8\text{ V}}$. All four plot points are well defined on Fig. 5.16 with the complete transfer curve.

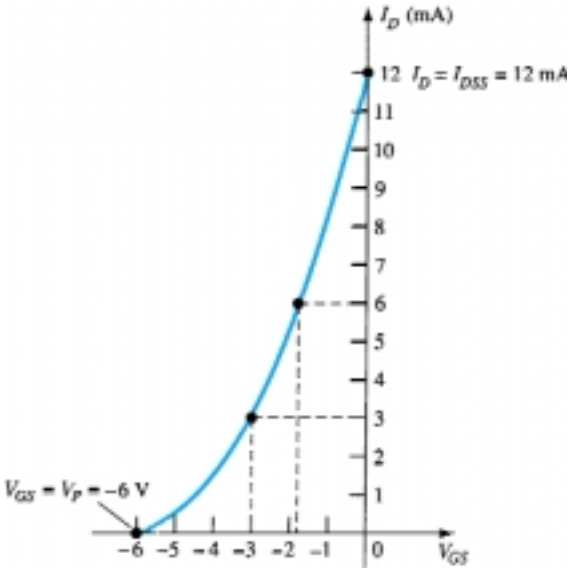


Figure 5.16 Transfer curve for Example 5.1.

For *p*-channel devices Shockley's equation (5.3) can still be applied exactly as it appears. In this case, both V_P and V_{GS} will be positive and the curve will be the mirror image of the transfer curve obtained with an *n*-channel and the same limiting values.

Sketch the transfer curve for a *p*-channel device with $I_{DSS} = 4$ mA and $V_P = 3$ V. *EXAMPLE 5.2*

Solution

At $V_{GS} = V_P/2 = 3\text{ V}/2 = \mathbf{1.5\text{ V}}$, $I_D = I_{DSS}/4 = 4\text{ mA}/4 = \mathbf{1\text{ mA}}$. At $I_D = I_{DSS}/2 = 4\text{ mA}/2 = \mathbf{2\text{ mA}}$, $V_{GS} = 0.3V_P = 0.3(3\text{ V}) = \mathbf{0.9\text{ V}}$. Both plot points appear in Fig. 5.17 along with the points defined by I_{DSS} and V_P.

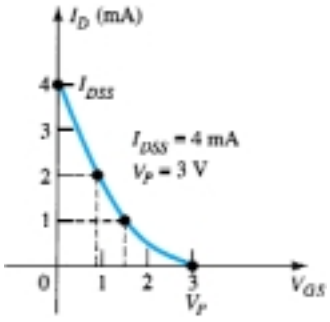


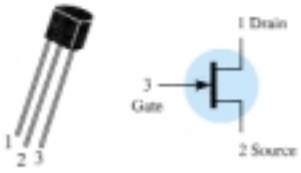
Figure 5.17 Transfer curve for the *p*-channel device of Example 5.2.

5.4 SPECIFICATION SHEETS (JFETs)

Although the general content of specification sheets may vary from the absolute minimum to an extensive display of graphs and charts, there are a few fundamental parameters that will be provided by all manufacturers. A few of the most important are discussed in the following paragraphs. The specification sheet for the 2N5457 *n*-channel JFET as provided by Motorola is provided as Fig. 5.18.

2N5457

CASE 29-04, STYLE 5
TO-92 (TO-226AA)



JFETs
GENERAL PURPOSE
N-CHANNEL—DEPLETION

Refer to 2N4220 for graphs.

MAXIMUM RATINGS

Rating	Symbol	Value	Unit
Drain-Source Voltage	V_{DS}	25	Vdc
Drain-Gate Voltage	V_{DG}	25	Vdc
Reverse Gate-Source Voltage	V_{GSR}	–25	Vdc
Gate Current	I_G	10	mAdc
Total Device Dissipation @ T_A = 25°C Derate above 25°C	P_D	310 2.82	mW mW/°C
Junction Temperature Range	T_J	125	°C
Storage Channel Temperature Range	T_{stg}	–65 to +150	°C

ELECTRICAL CHARACTERISTICS (T_A = 25°C unless otherwise noted)

Characteristic		Symbol	Min	Typ	Max	Unit
OFF CHARACTERISTICS						
Gate-Source Breakdown Voltage (I_G = –10 μAdc, V_{DS} = 0)		$V_{(BR)GSS}$	–25	–	–	Vdc
Gate Reverse Current (V_{GS} = –15 Vdc, V_{DS} = 0) (V_{GS} = –15 Vdc, V_{DS} = 0, T_A = 100°C)		I_{GSS}	– –	– –	–1.0 –200	nAdc
Gate Source Cutoff Voltage (V_{DS} = 15 Vdc, I_D = 10 nAdc)	2N5457	$V_{GS(off)}$	–0.5	–	–6.0	Vdc
Gate Source Voltage (V_{DS} = 15 Vdc, I_D = 100 μAdc)	2N5457	V_{GS}	–	–2.5	–	Vdc
ON CHARACTERISTICS						
Zero-Gate-Voltage Drain Current* (V_{DS} = 15 Vdc, V_{GS} = 0)	2N5457	I_{DSS}	1.0	3.0	5.0	mAdc
SMALL-SIGNAL CHARACTERISTICS						
Forward Transfer Admittance Common Source* (V_{DS} = 15 Vdc, V_{GS} = 0, f = 1.0 kHz)	2N5457	$\|y_{fs}\|$	1000	–	5000	μmhos
Output Admittance Common Source* (V_{DS} = 15 Vdc, V_{GS} = 0, f = 1.0 kHz)		$\|y_{os}\|$	–	10	50	μmhos
Input Capacitance (V_{DS} = 15 Vdc, V_{GS} = 0, f = 1.0 MHz)		C_{iss}	–	4.5	7.0	pF
Reverse Transfer Capacitance (V_{DS} = 15 Vdc, V_{GS} = 0, f = 1.0 MHz)		C_{rss}	–	1.5	3.0	pF

*Pulse Test: Pulse Width ≤ 630 ms; Duty Cycle ≤ 10%

Figure 5.18 2N5457 Motorola *n*-channel JFET.

Maximum Ratings

The maximum rating list usually appears at the beginning of the specification sheet, with the maximum voltages between specific terminals, maximum current levels, and the maximum power dissipation level of the device. The specified maximum levels for V_{DS} and V_{DG} must not be exceeded at any point in the design operation of the device. The applied source V_{DD} can exceed these levels, but the actual level of voltage between these terminals must never exceed the level specified. Any good design will

try to avoid these levels by a good margin of safety. The term *reverse* in V_{GSR} defines the maximum voltage with the source positive with respect to the gate (as normally biased for an *n*-channel device) before breakdown will occur. On some specification sheets it is referred to as BV_{DSS}—the *B*reakdown *V*oltage with the *D*rain-*S*ource *S*horted ($V_{DS} = 0$ V). Although normally designed to operate with $I_G = 0$ mA, if *forced* to accept a gate current it could withstand 10 mA before damage would occur. The total device dissipation at 25°C (room temperature) is the maximum power the device can dissipate under normal operating conditions and is defined by

$$P_D = V_{DS}I_D \tag{5.9}$$

Note the similarity in format with the maximum power dissipation equation for the BJT transistor.

The derating factor is discussed in detail in Chapter 3, but for the moment recognize that the 2.82 mW/°C rating reveals that the dissipation rating *decreases* by 2.82 mW for each *increase* in temperature of 1°C above 25°C.

Electrical Characteristics

The electrical characteristics include the level of V_P in the OFF CHARACTERISTICS and I_{DSS} in the ON CHARACTERISTICS. In this case $V_P = V_{GS(off)}$ has a range from -0.5 to -6.0 V and I_{DSS} from 1 to 5 mA. The fact that both will vary from device to device with the same nameplate identification must be considered in the design process. The other quantities are defined under conditions appearing in parentheses. The small-signal characteristics are discussed in Chapter 9.

Case Construction and Terminal Identification

This particular JFET has the appearance provided on the specification sheet of Fig. 5.18. The terminal identification is also provided directly under the figure. JFETs are also available in top-hat containers, as shown in Fig. 5.19 with its terminal identification.

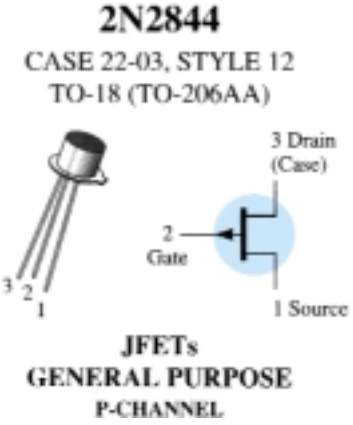


Figure 5.19 Top-hat container and terminal identification for a *p*-channel JFET.

Operating Region

The specification sheet and the curve defined by the pinch-off levels at each level of V_{GS} define the region of operation for linear amplification on the drain characteristics as shown in Fig. 5.20. The ohmic region defines the minimum permissible values of V_{DS} at each level of V_{GS}, and $V_{DS_{max}}$ specifies the maximum value for this pa-

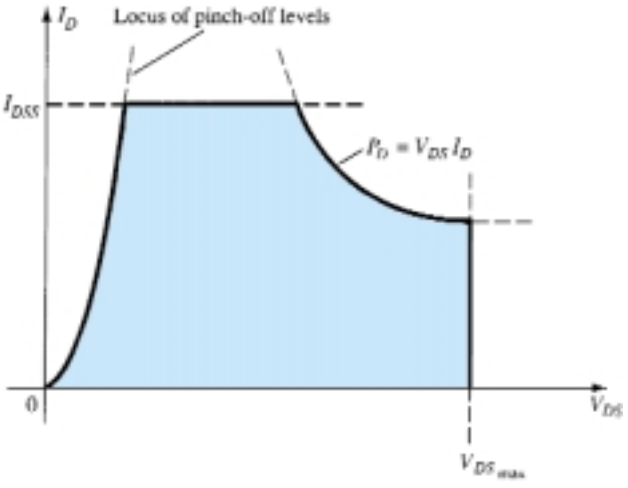


Figure 5.20 Normal operating region for linear amplifier design.

rameter. The saturation current I_{DSS} is the maximum drain current, and the maximum power dissipation level defines the curve drawn in the same manner as described for BJT transistors. The resulting shaded region is the normal operating region for amplifier design.

5.5 INSTRUMENTATION

Recall from Chapter 3 that hand-held instruments are available to measure the level of β_{dc} for the BJT transistor. Similar instrumentation is not available to measure the levels of I_{DSS} and V_P. However, the curve tracer introduced for the BJT transistor can also display the drain characteristics of the JFET transistor through a proper setting of the various controls. The vertical scale (in milliamperes) and the horizontal scale (in volts) have been set to provide a full display of the characteristics, as shown in Fig. 5.21. For the JFET of Fig. 5.21, each vertical division (in centimeters) reflects a 1-mA change in I_C while each horizontal division has a value of 1 V. The step voltage is 500 mV/step (0.5 V/step), revealing that the top curve is defined by $V_{GS} = 0$ V and the next curve down -0.5 V for the *n*-channel device. Using the same step voltage the next curve is -1 V, then -1.5 V, and finally -2 V. By drawing a line from the top curve over to the I_D axis, the level of I_{DSS} can be estimated to be about 9 mA. The level of V_P can be estimated by noting the V_{GS} value of the bottom curve and taking into account the shrinking distance between curves as V_{GS} becomes more and more negative. In this case, V_P is certainly more negative than -2 V and perhaps V_P is close to -2.5 V. However, keep in mind that the V_{GS} curves contract very quickly as they approach the cutoff condition, and perhaps $V_P = -3$ V is a better choice. It

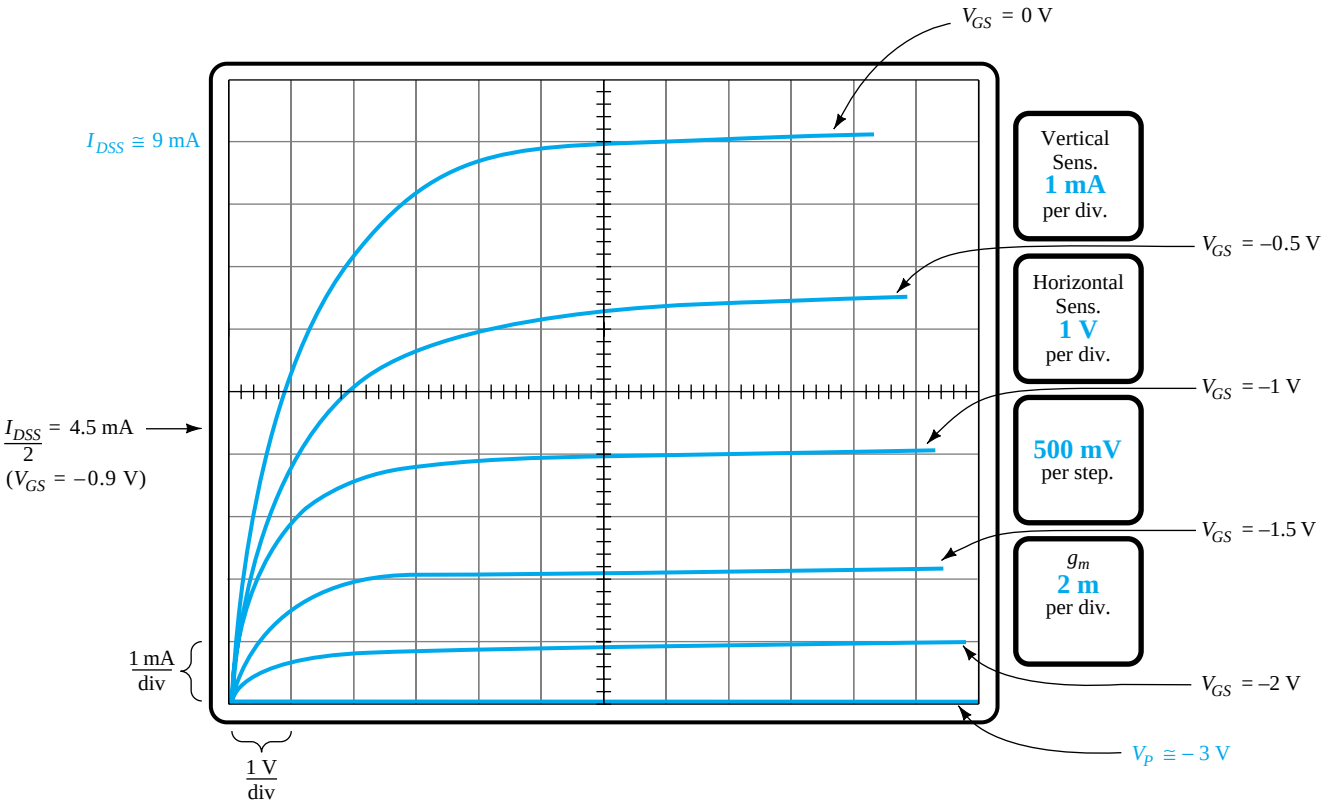


Figure 5.21 Drain characteristics for a 2N4416 JFET transistor as displayed on a curve tracer.

should also be noted that the step control is set for a 5-step display, limiting the displayed curves to $V_{GS} = 0$, -0.5, -1, -1.5, and -2 V. If the step control had been increased to 10, the voltage per step could be reduced to 250 mV = 0.25 V and the curve for $V_{GS} = -2.25$ V would have been included as well as an additional curve between each step of Fig. 5.21. The $V_{GS} = -2.25$ V curve would reveal how quickly the curves are closing in on each other for the same step voltage. Fortunately, the level of V_P can be estimated to a reasonable degree of accuracy simply by applying a condition appearing in Table 5.1. That is, when $I_D = I_{DSS}/2$, then $V_{GS} = 0.3V_P$. For the characteristics of Fig. 5.21, $I_D = I_{DSS}/2 = 9\text{ mA}/2 = 4.5$ mA, and as visible from Fig. 5.21 the corresponding level of V_{GS} is about -0.9 V. Using this information we find that $V_P = V_{GS}/0.3 = -0.9\text{ V}/0.3 = -3$ V, which will be our choice for this device. Using this value we find that at $V_{GS} = -2$ V,

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

$$= 9\text{ mA}\left(1 - \frac{-2\text{ V}}{-3\text{ V}}\right)^2$$

$$\cong 1\text{ mA}$$

as supported by Fig. 5.21.

At $V_{GS} = -2.5$ V, Shockley's equation will result in $I_D = 0.25$ mA, with $V_P = -3$ V clearly revealing how quickly the curves contract near V_P. The importance of the parameter g_m and how it is determined from the characteristics of Fig. 5.21 are described in Chapter 8 when small-signal ac conditions are examined.

5.6 IMPORTANT RELATIONSHIPS

A number of important equations and operating characteristics have been introduced in the last few sections that are of particular importance for the analysis to follow for the dc and ac configurations. In an effort to isolate and emphasize their importance, they are repeated below next to a corresponding equation for the BJT transistor. The JFET equations are defined for the configuration of Fig. 5.22a, while the BJT equations relate to Fig. 5.22b.

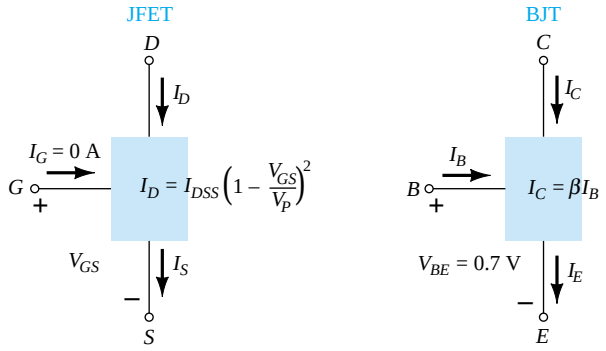


Figure 5.22 (a) JFET versus (b) BJT.

JFET		*BJT*
$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$	$\Leftrightarrow$	$I_C = \beta I_B$
$I_D = I_S$	$\Leftrightarrow$	$I_C \cong I_E$
$I_G \cong 0$ A	$\Leftrightarrow$	$V_{BE} \cong 0.7$ V

(5.10)

A clear understanding of the impact of each of the equations above is sufficient background to approach the most complex of dc configurations. Recall that $V_{BE} = 0.7$ V was often the key to initiating an analysis of a BJT configuration. Similarly, the condition $I_G = 0$ A is often the starting point for the analysis of a JFET configuration. For the BJT configuration, I_B is normally the first parameter to be determined. For the JFET, it is normally V_{GS}. The number of similarities between the analysis of BJT and JFET dc configurations will become quite apparent in Chapter 6.

5.7 DEPLETION-TYPE MOSFET

As noted in the chapter introduction, there are two types of FETs: JFETs and MOSFETs. MOSFETs are further broken down into *depletion type* and *enhancement type.* The terms *depletion* and *enhancement* define their basic mode of operation, while the label MOSFET stands for *m*etal-*o*xide-semiconductor-*f*ield-*e*ffect *t*ransistor. Since there are differences in the characteristics and operation of each type of MOSFET, they are covered in separate sections. In this section we examine the depletion-type MOSFET, which happens to have characteristics similar to those of a JFET between cutoff and saturation at I_{DSS} but then has the added feature of characteristics that extend into the region of opposite polarity for V_{GS}.

Basic Construction

The basic construction of the *n*-channel depletion-type MOSFET is provided in Fig. 5.23. A slab of *p*-type material is formed from a silicon base and is referred to as the *substrate.* It is the foundation upon which the device will be constructed. In some cases the substrate is internally connected to the source terminal. However, many discrete devices provide an additional terminal labeled *SS*, resulting in a four-terminal device, such as that appearing in Fig. 5.23. The source and drain terminals are connected through metallic contacts to *n*-doped regions linked by an *n*-channel as shown in the figure. The gate is also connected to a metal contact surface but remains insulated from the *n*-channel by a very thin silicon dioxide (SiO_2) layer. SiO_2 is a particular type of insulator referred to as a *dielectric* that sets up opposing (as revealed by

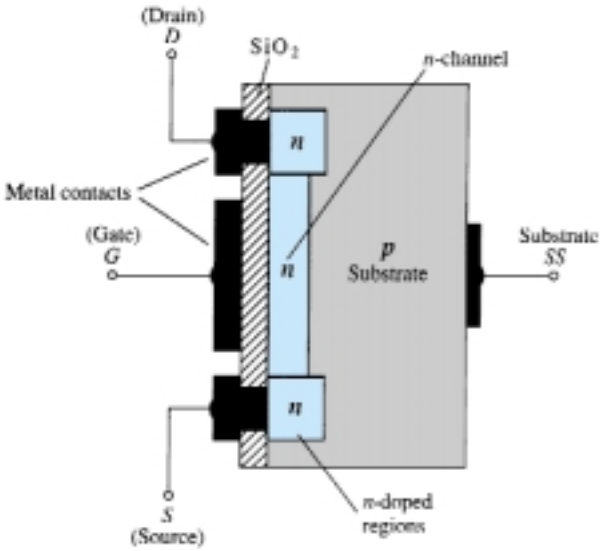


Figure 5.23 *n*-Channel depletion-type MOSFET.

the prefix *di-*) electric fields within the dielectric when exposed to an externally applied field. The fact that the SiO_2 layer is an insulating layer reveals the following fact:

There is no direct electrical connection between the gate terminal and the channel of a MOSFET.

In addition:

It is the insulating layer of SiO_2 in the MOSFET construction that accounts for the very desirable high input impedance of the device.

In fact, the input resistance of a MOSFET is often that of the typical JFET, even though the input impedance of most JFETs is sufficiently high for most applications. The very high input impedance continues to fully support the fact that the gate current (I_G) is essentially zero amperes for dc-biased configurations.

The reason for the label metal-oxide-semiconductor FET is now fairly obvious: *metal* for the drain, source, and gate connections to the proper surface—in particular, the gate terminal and the control to be offered by the surface area of the contact, the *oxide* for the silicon dioxide insulating layer, and the *semiconductor* for the basic structure on which the *n*- and *p*-type regions are diffused. The insulating layer between the gate and channel has resulted in another name for the device: *insulated-gate FET* or *IGFET,* although this label is used less and less in current literature.

Basic Operation and Characteristics

In Fig. 5.24 the gate-to-source voltage is set to zero volts by the direct connection from one terminal to the other, and a voltage V_{DS} is applied across the drain-to-source terminals. The result is an attraction for the positive potential at the drain by the *free* electrons of the *n*-channel and a current similar to that established through the channel of the JFET. In fact, the resulting current with $V_{GS} = 0$ V continues to be labeled I_{DSS}, as shown in Fig. 5.25.

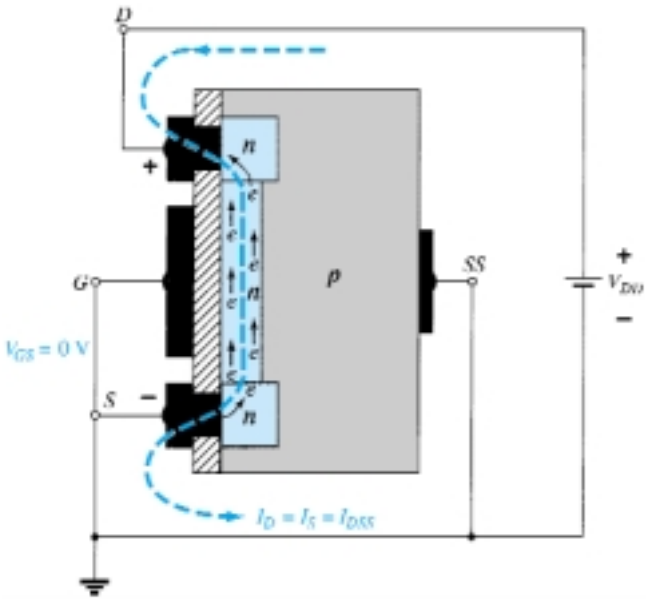


Figure 5.24 *n*-Channel depletion-type MOSFET with $V_{GS} = 0$ V and an applied voltage V_{DD}.

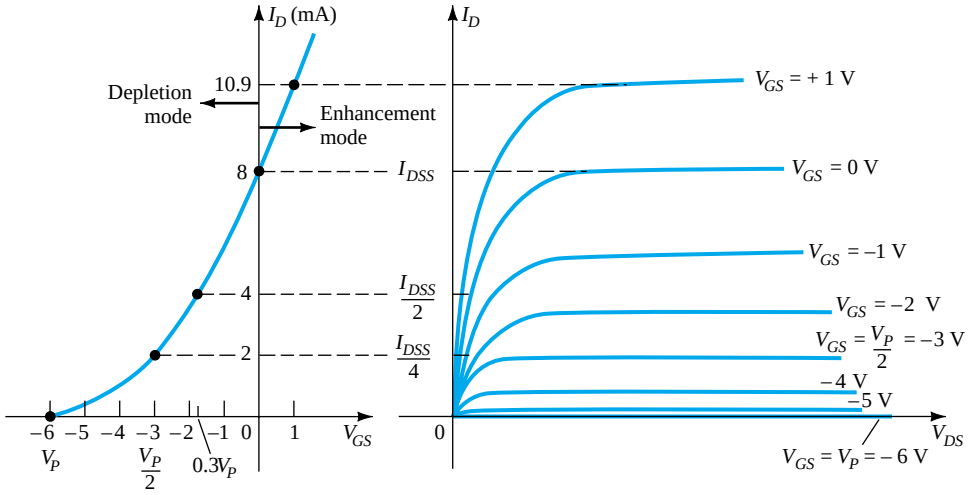


Figure 5.25 Drain and transfer characteristics for an *n*-channel depletion-type MOSFET.

In Fig. 5.26, V_{GS} has been set at a negative voltage such as -1 V. The negative potential at the gate will tend to pressure electrons toward the *p*-type substrate (like charges repel) and attract holes from the *p*-type substrate (opposite charges attract) as shown in Fig. 5.26. Depending on the magnitude of the negative bias established by V_{GS}, a level of recombination between electrons and holes will occur that will reduce the number of free electrons in the *n*-channel available for conduction. The more negative the bias, the higher the rate of recombination. The resulting level of drain current is therefore reduced with increasing negative bias for V_{GS} as shown in Fig. 5.25 for $V_{GS} = -1$ V, -2 V, and so on, to the pinch-off level of -6 V. The resulting levels of drain current and the plotting of the transfer curve proceeds exactly as described for the JFET.

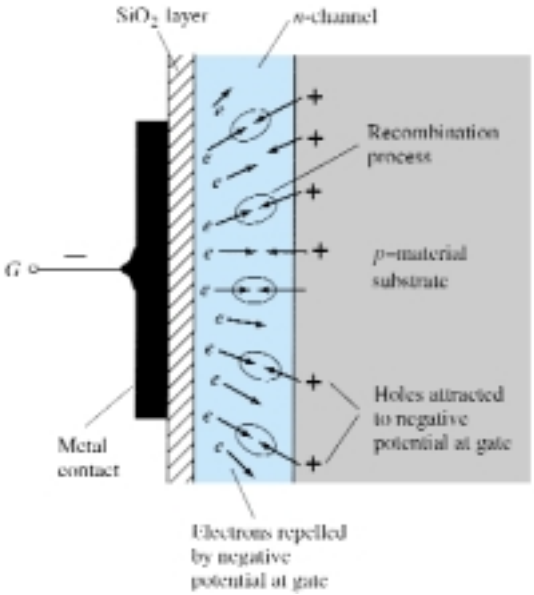


Figure 5.26 Reduction in free carriers in channel due to a negative potential at the gate terminal.

For positive values of V_{GS}, the positive gate will draw additional electrons (free carriers) from the *p*-type substrate due to the reverse leakage current and establish new carriers through the collisions resulting between accelerating particles. As the gate-to-source voltage continues to increase in the positive direction, Fig. 5.25 reveals that the drain current will increase at a rapid rate for the reasons listed above. The

vertical spacing between the $V_{GS} = 0$ V and $V_{GS} = +1$ V curves of Fig. 5.25 is a clear indication of how much the current has increased for the 1-V change in V_{GS}. Due to the rapid rise, the user must be aware of the maximum drain current rating since it could be exceeded with a positive gate voltage. That is, for the device of Fig. 5.25, the application of a voltage $V_{GS} = +4$ V would result in a drain current of 22.2 mA, which could possibly exceed the maximum rating (current or power) for the device. As revealed above, the application of a positive gate-to-source voltage has "enhanced" the level of free carriers in the channel compared to that encountered with $V_{GS} = 0$ V. For this reason the region of positive gate voltages on the drain or transfer characteristics is often referred to as the *enhancement region*, with the region between cutoff and the saturation level of I_{DSS} referred to as the *depletion region*.

It is particularly interesting and helpful that Shockley's equation will continue to be applicable for the depletion-type MOSFET characteristics in both the depletion and enhancement regions. For both regions, it is simply necessary that the proper sign be included with V_{GS} in the equation and the sign be carefully monitored in the mathematical operations.

EXAMPLE 5.3

Sketch the transfer characteristics for an *n*-channel depletion-type MOSFET with $I_{DSS} = 10$ mA and $V_P = -4$ V.

Solution

$$\text{At } V_{GS} = 0 \text{ V}, \qquad I_D = I_{DSS} = 10 \text{ mA}$$

$$V_{GS} = V_P = -4 \text{ V}, \qquad I_D = 0 \text{ mA}$$

$$V_{GS} = \frac{V_P}{2} = \frac{-4 \text{ V}}{2} = -2 \text{ V}, \qquad I_D = \frac{I_{DSS}}{4} = \frac{10 \text{ mA}}{4} = 2.5 \text{ mA}$$

and at $I_D = \dfrac{I_{DSS}}{2}$, $\qquad V_{GS} = 0.3V_P = 0.3(-4 \text{ V}) = -1.2 \text{ V}$

all of which appear in Fig. 5.27.

Before plotting the positive region of V_{GS}, keep in mind that I_D increases very rapidly with increasing positive values of V_{GS}. In other words, be conservative with the choice of values to be substituted into Shockley's equation. In this case, we will try $+1$ V as follows:

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

$$= 10 \text{ mA}\left(1 - \frac{+1 \text{ V}}{-4 \text{ V}}\right)^2 = 10 \text{ mA}(1 + 0.25)^2 = 10 \text{ mA}(1.5625)$$

$$\cong 15.63 \text{ mA}$$

which is sufficiently high to finish the plot.

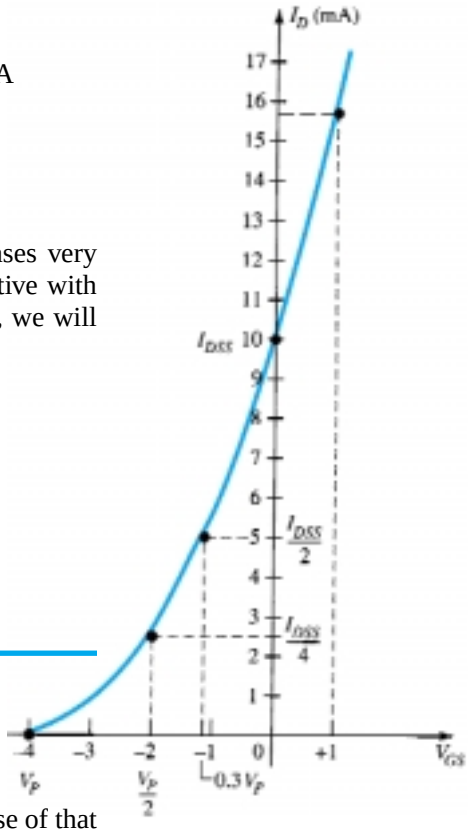


Figure 5.27 Transfer characteristics for an *n*-channel depletion-type MOSFET with $I_{DSS} = 10$ mA and $V_P = -4$ V.

p-Channel Depletion-Type MOSFET

The construction of a *p*-channel depletion-type MOSFET is exactly the reverse of that appearing in Fig. 5.23. That is, there is now an *n*-type substrate and a *p*-type channel, as shown in Fig. 5.28a. The terminals remain as identified, but all the voltage polarities and the current directions are reversed, as shown in the same figure. The drain characteristics would appear exactly as in Fig. 5.25 but with V_{DS} having negative val-

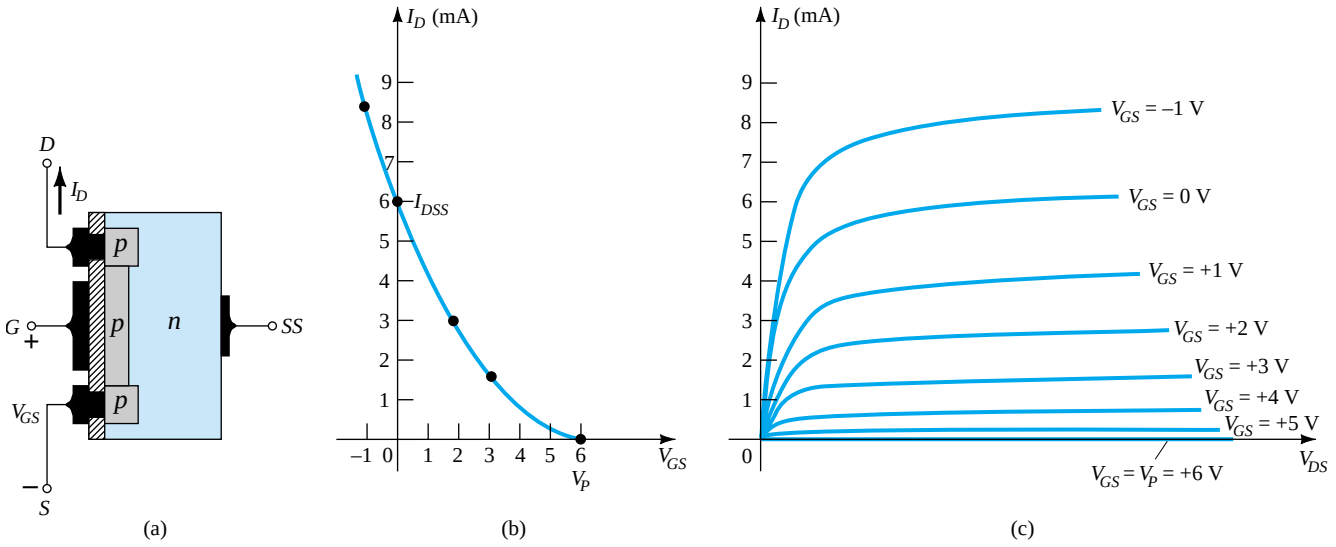


Figure 5.28 *p*-Channel depletion-type MOSFET with $I_{DSS} = 6$ mA and $V_P = +6$ V.

ues, I_D having positive values as indicated (since the defined direction is now reversed), and V_{GS} having the opposite polarities as shown in Fig. 5.28c. The reversal in V_{GS} will result in a mirror image (about the I_D axis) for the transfer characteristics as shown in Fig. 5.28b. In other words, the drain current will increase from cutoff at $V_{GS} = V_P$ in the positive V_{GS} region to I_{DSS} and then continue to increase for increasingly negative values of V_{GS}. Shockley's equation is still applicable and requires simply placing the correct sign for both V_{GS} and V_P in the equation.

Symbols, Specification Sheets, and Case Construction

The graphic symbols for an *n*- and *p*-channel depletion-type MOSFET are provided in Fig. 5.29. Note how the symbols chosen try to reflect the actual construction of the device. The lack of a direct connection (due to the gate insulation) between the gate and channel is represented by a space between the gate and the other terminals of the symbol. The vertical line representing the channel is connected between the drain and source and is "supported" by the substrate. Two symbols are provided for each type of channel to reflect the fact that in some cases the substrate is externally available while in others it is not. For most of the analysis to follow in Chapter 6, the substrate and source will be connected and the lower symbols will be employed.

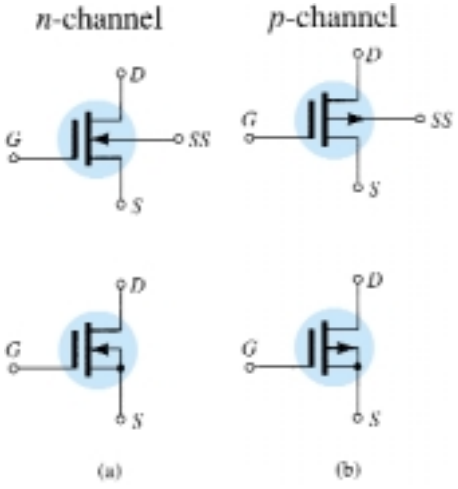
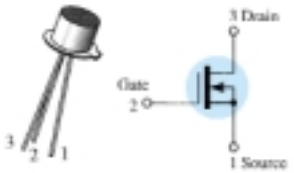


Figure 5.29 Graphic symbols for (a) *n*-channel depletion-type MOSFETs and (b) *p*-channel depletion-type MOSFETs.

The device appearing in Fig. 5.30 has three terminals, with the terminal identification appearing in the same figure. The specification sheet for a depletion-type MOSFET is similar to that of a JFET. The levels of V_P and I_{DSS} are provided along with a list of maximum values and typical "on" and "off" characteristics. In addition, how-

2N3797

CASE 22-03, STYLE 2
TO-18 (TO-206AA)



MOSFETs
LOW POWER AUDIO

N-CHANNEL – DEPLETION

MAXIMUM RATINGS

Rating	Symbol	Value	Unit
Drain–Source Voltage 2N3797	V_{DS}	20	Vdc
Gate–Source Voltage	V_{GS}	±10	Vdc
Drain Current	I_D	20	mAdc
Total Device Dissipation @ T_A = 25°C Derate above 25°C	P_D	200 1.14	mW mW/°C
Junction Temperature Range	T_J	+175	°C
Storage Channel Temperature Range	T_{stg}	–65 to +200	°C

ELECTRICAL CHARACTERISTICS (T_A = 25°C unless otherwise noted)

Characteristic		Symbol	Min	Typ	Max	Unit
OFF CHARACTERISTICS						
Drain Source Breakdown Voltage (V_{GS} = –7.0 V, I_D = 5.0 µA)	2N3797	$V_{(BR)DSX}$	20	25	–	Vdc
Gate Reverse Current (1) (V_{GS} = –10 V, V_{DS} = 0) (V_{GS} = –10 V, V_{DS} = 0, T_A = 150°C)		I_{GSS}	– –	– –	1.0 200	pAdc
Gate Source Cutoff Voltage (I_D = 2.0 µA, V_{DS} = 10 V)	2N3797	$V_{GS(off)}$	–	–5.0	–7.0	Vdc
Drain-Gate Reverse Current (1) (V_{DG} = 10 V, I_S = 0)		I_{DGO}	–	–	1.0	pAdc
ON CHARACTERISTICS						
Zero-Gate-Voltage Drain Current (V_{DS} = 10 V, V_{GS} = 0)	2N3797	I_{DSS}	2.0	2.9	6.0	mAdc
On-State Drain Current (V_{DS} = 10 V, V_{GS} = +3.5 V)	2N3797	$I_{D(on)}$	9.0	14	18	mAdc
SMALL-SIGNAL CHARACTERISTICS						
Forward Transfer Admittance (V_{DS} = 10 V, V_{GS} = 0, f = 1.0 kHz)	2N3797	$\lvert Y_{fs}\rvert$	1500	2300	3000	µmhos
(V_{DS} = 10 V, V_{GS} = 0, f = 1.0 MHz)	2N3797		1500	–	–	
Output Admittance (I_{DS} = 10 V, V_{GS} = 0, f = 1.0 kHz)	2N3797	$\lvert Y_{os}\rvert$	–	27	60	µmhos
Input Capacitance (V_{DS} = 10 V, V_{GS} = 0, f = 1.0 MHz)	2N3797	C_{iss}	–	6.0	8.0	pF
Reverse Transfer Capacitance (V_{DS} = 10 V, V_{GS} = 0, f = 1.0 MHz)		C_{rss}	–	0.5	0.8	pF
FUNCTIONAL CHARACTERISTICS						
Noise Figure (V_{DS} = 10 V, V_{GS} = 0, f = 1.0 kHz, R_S = 3 megohms)		NF	–	3.8	–	dB

(1) This value of current includes both the FET leakage current as well as the leakage current associated with the test socket and fixture when measured under best attainable conditions.

Figure 5.30 2N3797 Motorola *n*-channel depletion-type MOSFET.

ever, since I_D can extend beyond the I_{DSS} level, another point is normally provided that reflects a typical value of I_D for some positive voltage (for an *n*-channel device). For the unit of Fig. 5.30, I_D is specified as $I_{D(on)} = 9$ mA dc, with $V_{DS} = 10$ V and $V_{GS} = 3.5$ V.

5.8 ENHANCEMENT-TYPE MOSFET

Although there are some similarities in construction and mode of operation between depletion-type and enhancement-type MOSFETs, the characteristics of the enhancement-type MOSFET are quite different from anything obtained thus far. The transfer curve is not defined by Shockley's equation, and the drain current is now cut off until the gate-to-source voltage reaches a specific magnitude. In particular, current control in an *n*-channel device is now effected by a positive gate-to-source voltage rather than the range of negative voltages encountered for *n*-channel JFETs and *n*-channel depletion-type MOSFETs.

Basic Construction

The basic construction of the *n*-channel enhancement-type MOSFET is provided in Fig. 5.31. A slab of *p*-type material is formed from a silicon base and is again referred to as the substrate. As with the depletion-type MOSFET, the substrate is sometimes internally connected to the source terminal, while in other cases a fourth lead is made available for external control of its potential level. The source and drain terminals are again connected through metallic contacts to *n*-doped regions, but note in Fig. 5.31 the absence of a channel between the two *n*-doped regions. This is the primary difference between the construction of depletion-type and enhancement-type MOSFETs—the absence of a channel as a constructed component of the device. The SiO_2 layer is still present to isolate the gate metallic platform from the region between the drain and source, but now it is simply separated from a section of the *p*-type material. In summary, therefore, the construction of an enhancement-type MOSFET is quite similar to that of the depletion-type MOSFET, except for the absence of a channel between the drain and source terminals.

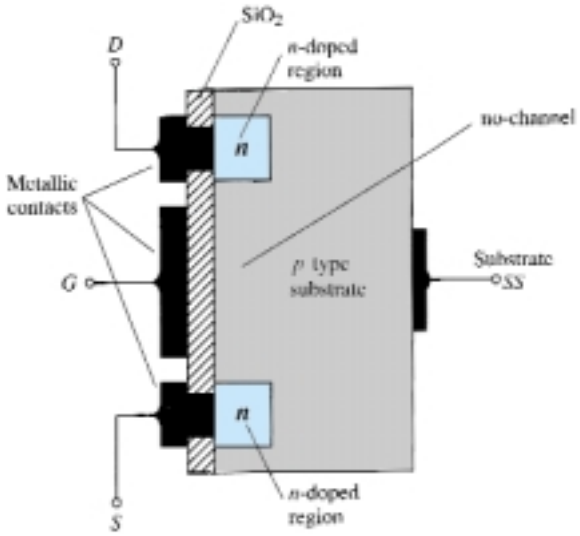


Figure 5.31 *n*-Channel enhancement-type MOSFET.

Basic Operation and Characteristics

If V_{GS} is set at 0 V and a voltage applied between the drain and source of the device of Fig. 5.31, the absence of an *n*-channel (with its generous number of free carriers) will result in a current of effectively zero amperes—quite different from the depletion-type MOSFET and JFET where $I_D = I_{DSS}$. It is not sufficient to have a large accumulation of carriers (electrons) at the drain and source (due to the *n*-doped regions) if a path fails to exist between the two. With V_{DS} some positive voltage, V_{GS} at 0 V, and terminal *SS* directly connected to the source, there are in fact two reverse-biased *p*-*n* junctions between the *n*-doped regions and the *p*-substrate to oppose any significant flow between drain and source.

In Fig. 5.32 both V_{DS} and V_{GS} have been set at some positive voltage greater than 0 V, establishing the drain and gate at a positive potential with respect to the source. The positive potential at the gate will pressure the holes (since like charges repel) in the *p*-substrate along the edge of the SiO_2 layer to leave the area and enter deeper regions of the *p*-substrate, as shown in the figure. The result is a depletion region near the SiO_2 insulating layer void of holes. However, the electrons in the *p*-substrate (the minority carriers of the material) will be attracted to the positive gate and accumulate in the region near the surface of the SiO_2 layer. The SiO_2 layer and its insulating qualities will prevent the negative carriers from being absorbed at the gate terminal. As V_{GS} increases in magnitude, the concentration of electrons near the SiO_2 surface increases until eventually the induced *n*-type region can support a measurable flow between drain and source. The level of V_{GS} that results in the significant increase in drain current is called the *threshold voltage* and is given the symbol V_T. On specification sheets it is referred to as $V_{GS(Th)}$, although V_T is less unwieldy and will be used in the analysis to follow. Since the channel is nonexistent with $V_{GS} = 0$ V and "enhanced" by the application of a positive gate-to-source voltage, this type of MOSFET is called an *enhancement-type MOSFET*. Both depletion- and enhancement-type MOSFETs have enhancement-type regions, but the label was applied to the latter since it is its only mode of operation.

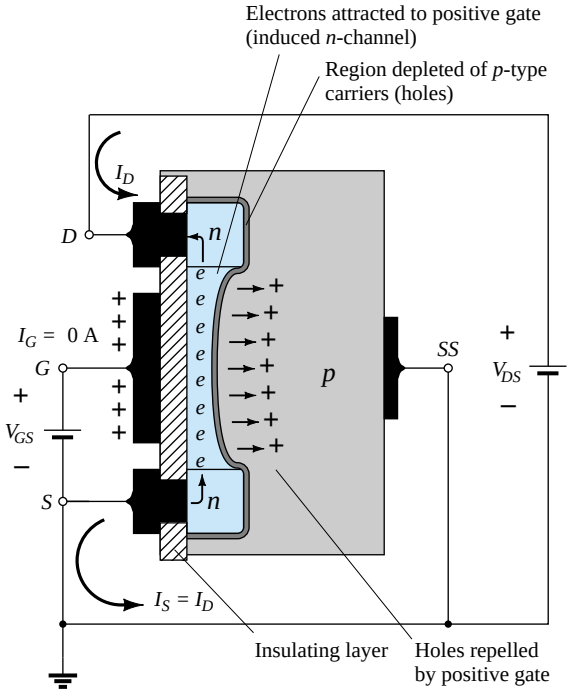


Figure 5.32 Channel formation in the *n*-channel enhancement-type MOSFET.

As V_{GS} is increased beyond the threshold level, the density of free carriers in the induced channel will increase, resulting in an increased level of drain current. However, if we hold V_{GS} constant and increase the level of V_{DS}, the drain current will eventually reach a saturation level as occurred for the JFET and depletion-type MOSFET. The leveling off of I_D is due to a pinching-off process depicted by the narrower channel at the drain end of the induced channel as shown in Fig. 5.33. Applying Kirchhoff's voltage law to the terminal voltages of the MOSFET of Fig. 5.33, we find that

$$V_{DG} = V_{DS} - V_{GS} \tag{5.11}$$

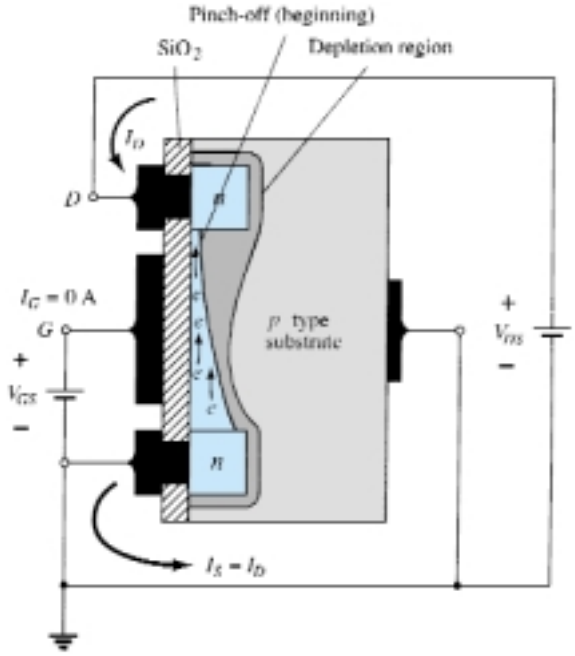


Figure 5.33 Change in channel and depletion region with increasing level of V_{DS} for a fixed value of V_{GS}.

If V_{GS} is held fixed at some value such as 8 V and V_{DS} is increased from 2 to 5 V, the voltage V_{DG} [by Eq. (5.11)] will drop from -6 to -3 V and the gate will become less and less positive with respect to the drain. This reduction in gate-to-drain voltage will in turn reduce the attractive forces for free carriers (electrons) in this region of the induced channel, causing a reduction in the effective channel width. Eventually, the channel will be reduced to the point of pinch-off and a saturation condition will be established as described earlier for the JFET and depletion-type MOSFET. In other words, any further increase in V_{DS} at the fixed value of V_{GS} will not affect the saturation level of I_D until breakdown conditions are encountered.

The drain characteristics of Fig. 5.34 reveal that for the device of Fig. 5.33 with $V_{GS} = 8$ V, saturation occurred at a level of $V_{DS} = 6$ V. In fact, the saturation level for V_{DS} is related to the level of applied V_{GS} by

$$V_{DS_{\text{sat}}} = V_{GS} - V_T \tag{5.12}$$

Obviously, therefore, for a fixed value of V_T, then the higher the level of V_{GS}, the more the saturation level for V_{DS}, as shown in Fig. 5.33 by the locus of saturation levels.

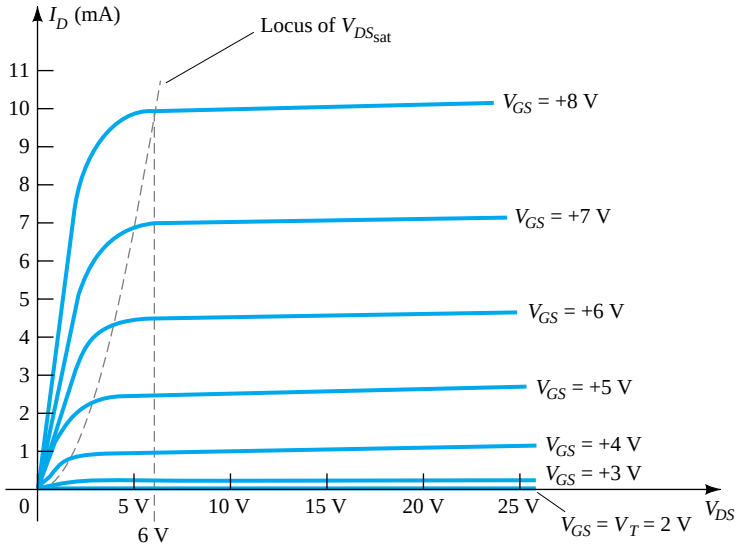


Figure 5.34 Drain characteristics of an *n*-channel enhancement-type MOSFET with $V_T = 2$ V and $k = 0.278 \times 10^{-3}$ A/V^2.

For the characteristics of Fig. 5.33 the level of V_T is 2 V, as revealed by the fact that the drain current has dropped to 0 mA. In general, therefore:

For values of V_{GS} less than the threshold level, the drain current of an enhancement-type MOSFET is 0 mA.

Figure 5.34 clearly reveals that as the level of V_{GS} increased from V_T to 8 V, the resulting saturation level for I_D also increased from a level of 0 to 10 mA. In addition, it is quite noticeable that the spacing between the levels of V_{GS} increased as the magnitude of V_{GS} increased, resulting in ever-increasing increments in drain current.

For levels of $V_{GS} > V_T$, the drain current is related to the applied gate-to-source voltage by the following nonlinear relationship:

$$I_D = k(V_{GS} - V_T)^2 \tag{5.13}$$

Again, it is the squared term that results in the nonlinear (curved) relationship between I_D and V_{GS}. The k term is a constant that is a function of the construction of the device. The value of k can be determined from the following equation [derived from Eq. (5.13)] where $I_{D(on)}$ and $V_{GS(on)}$ are the values for each at a particular point on the characteristics of the device.

$$k = \frac{I_{D(on)}}{(V_{GS(on)} - V_T)^2} \tag{5.14}$$

Substituting $I_{D(on)} = 10$ mA when $V_{GS(on)} = 8$ V from the characteristics of Fig. 5.34 yields

$$k = \frac{10 \text{ mA}}{(8 \text{ V} - 2 \text{ V})^2} = \frac{10 \text{ mA}}{(6 \text{ V})^2} = \frac{10 \text{ mA}}{36 \text{ V}^2}$$
$$= \mathbf{0.278 \times 10^{-3} \text{ A/V}^2}$$

and a general equation for I_D for the characteristics of Fig. 5.34 results in:

$$I_D = 0.278 \times 10^{-3}(V_{GS} - 2 \text{ V})^2$$

Substituting $V_{GS} = 4$ V, we find that

$$I_D = 0.278 \times 10^{-3}(4\text{ V} - 2\text{ V})^2 = 0.278 \times 10^{-3}(2)^2$$
$$= 0.278 \times 10^{-3}(4) = \mathbf{1.11\ mA}$$

as verified by Fig. 5.34. At $V_{GS} = V_T$, the squared term is 0 and $I_D = 0$ mA.

For the dc analysis of enhancement-type MOSFETs to appear in Chapter 6, the transfer characteristics will again be the characteristics to be employed in the graphical solution. In Fig. 5.35 the drain and transfer characteristics have been set side by side to describe the transfer process from one to the other. Essentially, it proceeds as introduced earlier for the JFET and depletion-type MOSFETs. In this case, however, it must be remembered that the drain current is 0 mA for $V_{GS} \leq V_T$. At this point a measurable current will result for I_D and will increase as defined by Eq. (5.13). Note that in defining the points on the transfer characteristics from the drain characteristics, only the saturation levels are employed, thereby limiting the region of operation to levels of V_{DS} greater than the saturation levels as defined by Eq. (5.12).

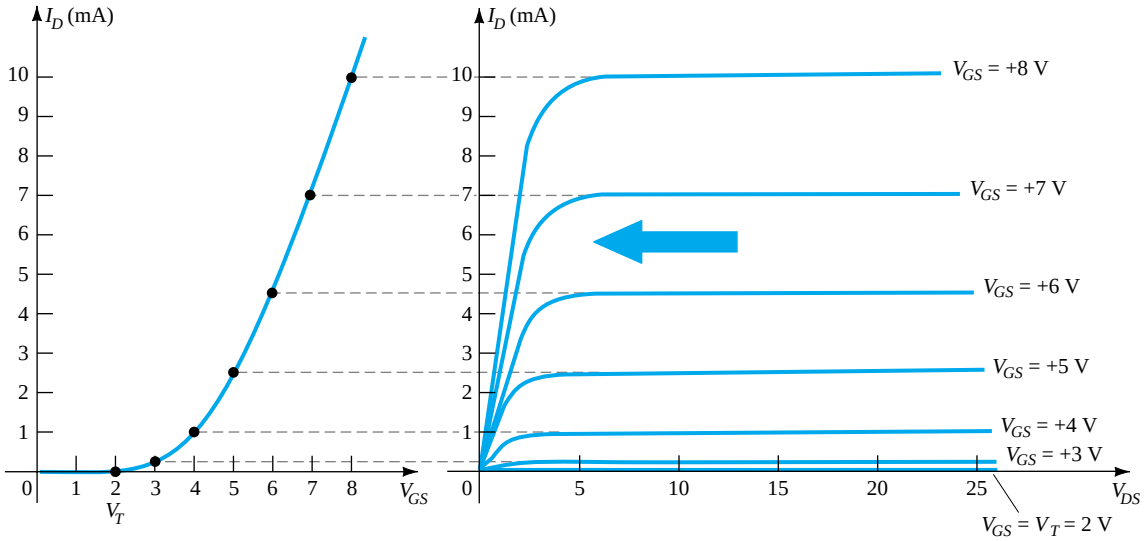


Figure 5.35 Sketching the transfer characteristics for an *n*-channel enhancement-type MOSFET from the drain characteristics.

The transfer curve of Fig. 5.35 is certainly quite different from those obtained earlier. For an *n*-channel (induced) device, it is now totally in the positive V_{GS} region and does not rise until $V_{GS} = V_T$. The question now surfaces as to how to plot the transfer characteristics given the levels of k and V_T as included below for a particular MOSFET:

$$I_D = 0.5 \times 10^{-3}(V_{GS} - 4\text{ V})^2$$

First, a horizontal line is drawn at $I_D = 0$ mA from $V_{GS} = 0$ V to $V_{GS} = 4$ V as shown in Fig. 5.36a. Next, a level of V_{GS} greater than V_T such as 5 V is chosen and substituted into Eq. (5.13) to determine the resulting level of I_D as follows:

$$I_D = 0.5 \times 10^{-3}(V_{GS} - 4\text{ V})^2$$
$$= 0.5 \times 10^{-3}(5\text{ V} - 4\text{ V})^2 = 0.5 \times 10^{-3}(1)^2$$
$$\mathbf{= 0.5\ mA}$$

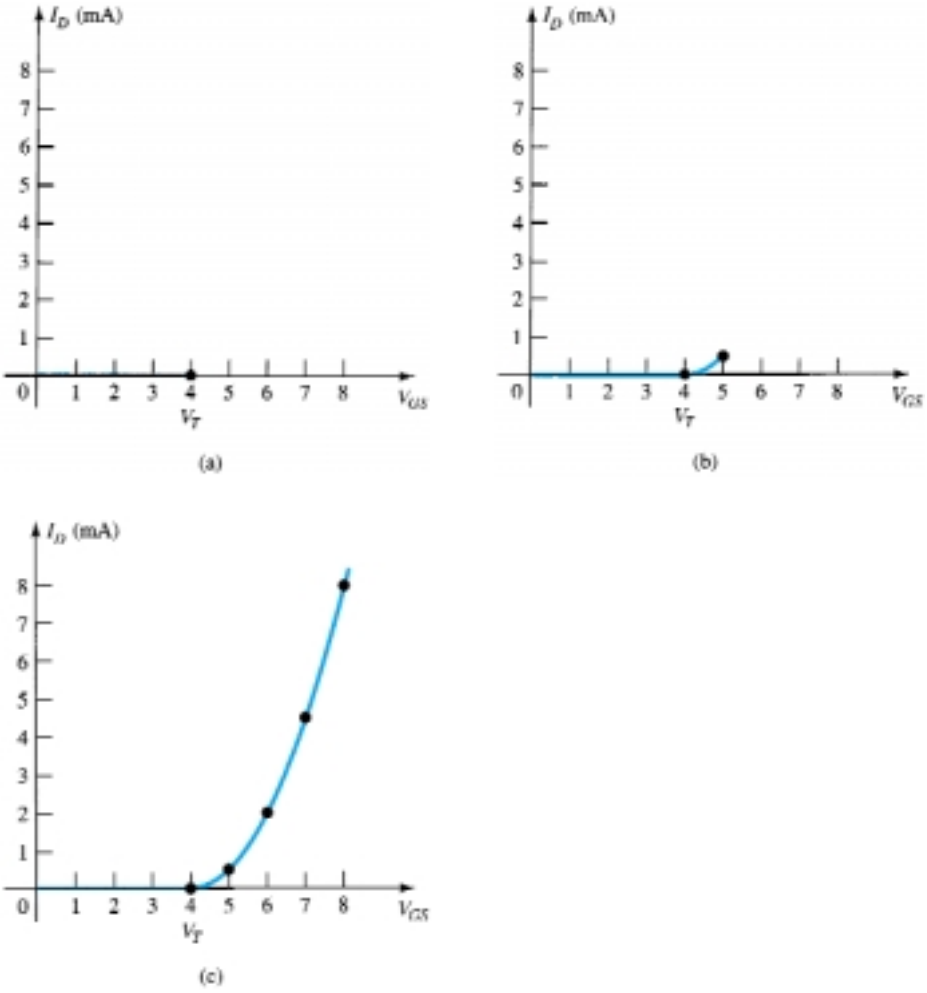


Figure 5.36 Plotting the transfer characteristics of an *n*-channel enhancement-type MOSFET with $k = 0.5 \times 10^{-3}$ A/V^2 and $V_T = 4$ V.

and a point on the plot is obtained as shown in Fig. 5.36b. Finally, additional levels of V_{GS} are chosen and the resulting levels of I_D obtained. In particular, at $V_{GS} = 6$, 7, and 8 V, the level of I_D is 2, 4.5, and 8 mA, respectively, as shown on the resulting plot of Fig. 5.36c.

p-Channel Enhancement-Type MOSFETs

The construction of a *p*-channel enhancement-type MOSFET is exactly the reverse of that appearing in Fig. 5.31, as shown in Fig. 5.37a. That is, there is now an *n*-type substrate and *p*-doped regions under the drain and source connections. The terminals remain as identified, but all the voltage polarities and the current directions are reversed. The drain characteristics will appear as shown in Fig. 5.37c, with increasing levels of current resulting from increasingly negative values of V_{GS}. The transfer characteristics will be the mirror image (about the I_D axis) of the transfer curve of Fig. 5.35, with I_D increasing with increasingly negative values of V_{GS} beyond V_T, as shown in Fig. 5.37b. Equations (5.11) through (5.14) are equally applicable to *p*-channel devices.

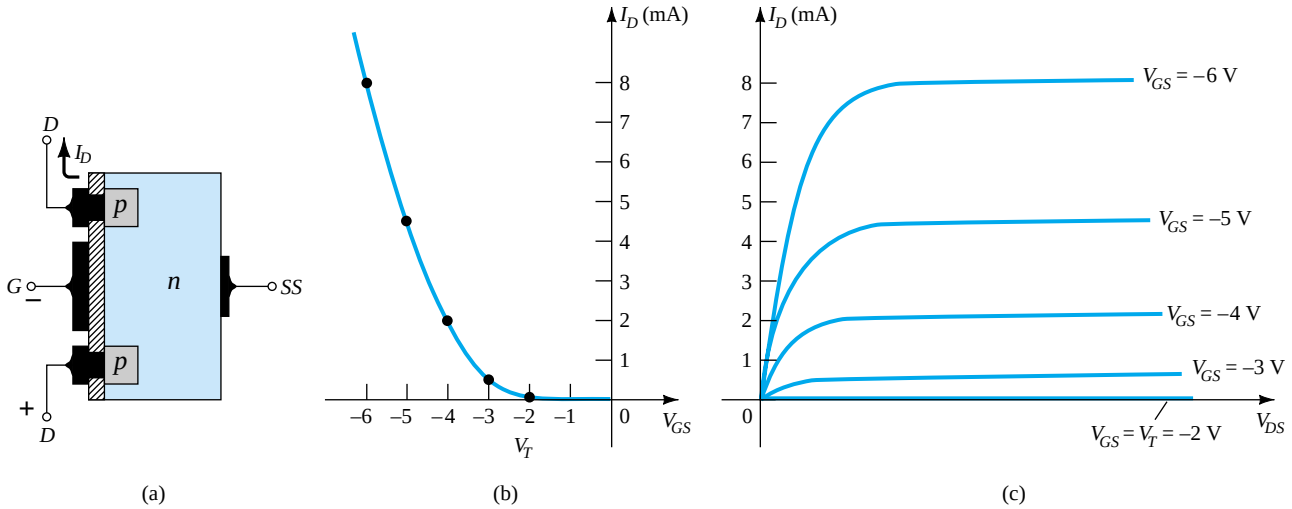


Figure 5.37 *p*-Channel enhancement-type MOSFET with $V_T = 2$ V and $k = 0.5 \times 10^{-3}$ A/V^2.

Symbols, Specification Sheets, and Case Construction

The graphic symbols for the *n*- and *p*-channel enhancement-type MOSFETs are provided as Fig. 5.38. Again note how the symbols try to reflect the actual construction of the device. The dashed line between drain and source was chosen to reflect the fact that a channel does not exist between the two under no-bias conditions. It is, in fact, the only difference between the symbols for the depletion-type and enhancement-type MOSFETs.

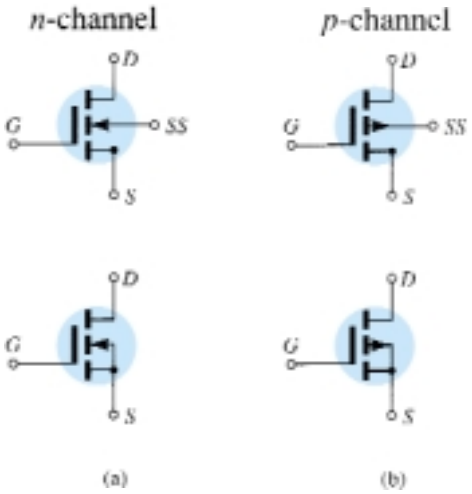
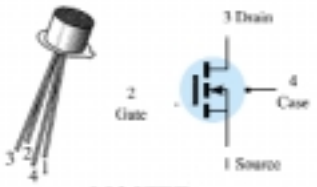


Figure 5.38 Symbols for (a) *n*-channel enhancement-type MOSFETs and (b) *p*-channel enhancement-type MOSFETs.

The specification sheet for a Motorola *n*-channel enhancement-type MOSFET is provided as Fig. 5.39. The case construction and terminal identification are provided next to the maximum ratings, which now include a maximum drain current of 30 mA dc. The specification sheet provides the level of I_{DSS} under "off" conditions, which is now simply 10 nA dc (at $V_{DS} = 10$ V and $V_{GS} = 0$ V) compared to the milliampere range for the JFET and depletion-type MOSFET. The threshold voltage is specified

2N4351

CASE 20-03, STYLE 2
TO-72 (TO-206AF)



MOSFET SWITCHING
N-CHANNEL – ENHANCEMENT

MAXIMUM RATINGS

Rating	Symbol	Value	Unit
Drain–Source Voltage	V_{DS}	25	Vdc
Drain–Gate Voltage	V_{DG}	30	Vdc
Gate–Source Voltage*	V_{GS}	30	Vdc
Drain Current	I_D	30	mAdc
Total Device Dissipation @ $T_A = 25°C$ Derate above 25°C	P_D	300 1.7	mW mW/°C
Junction Temperature Range	T_J	175	°C
Storage Temperature Range	T_{stg}	−65 to +175	°C

* Transient potentials of ± 75 Volt will not cause gate-oxide failure.

ELECTRICAL CHARACTERISTICS ($T_A = 25°C$ unless otherwise noted.)

Characteristic	Symbol	Min	Max	Unit
OFF CHARACTERISTICS				
Drain-Source Breakdown Voltage ($I_D = 10\ \mu A$, $V_{GS} = 0$)	$V_{(BR)DSX}$	25	–	Vdc
Zero-Gate-Voltage Drain Current ($V_{DS} = 10$ V, $V_{GS} = 0$) $T_A = 25°C$ $T_A = 150°C$	I_{DSS}	– –	10 10	nAdc μAdc
Gate Reverse Current ($V_{GS} = \pm 15$ Vdc, $V_{DS} = 0$)	I_{GSS}	–	± 10	pAdc
ON CHARACTERISTICS				
Gate Threshold Voltage ($V_{DS} = 10$ V, $I_D = 10\ \mu A$)	$V_{GS(Th)}$	1.0	5	Vdc
Drain-Source On-Voltage ($I_D = 2.0$ mA, $V_{GS} = 10$V)	$V_{DS(on)}$	–	1.0	V
On-State Drain Current ($V_{GS} = 10$ V, $V_{DS} = 10$ V)	$I_{D(on)}$	3.0	–	mAdc
SMALL-SIGNAL CHARACTERISTICS				
Forward Transfer Admittance ($V_{DS} = 10$ V, $I_D = 2.0$ mA, f = 1.0 kHz)	$\lvert y_{fs} \rvert$	1000	–	μmho
Input Capacitance ($V_{DS} = 10$ V, $V_{GS} = 0$, f = 140 kHz)	C_{iss}	–	5.0	pF
Reverse Transfer Capacitance ($V_{DS} = 0$, $V_{GS} = 0$, f = 140 kHz)	C_{rss}	–	1.3	pF
Drain-Substrate Capacitance ($V_{D(SUB)} = 10$ V, f = 140 kHz)	$C_{d(sub)}$	–	5.0	pF
Drain-Source Resistance ($V_{GS} = 10$ V, $I_D = 0$, f = 1.0 kHz)	$r_{ds(on)}$	–	300	ohms
SWITCHING CHARACTERISTICS				
Turn-On Delay (Fig. 5)	t_{d1}	–	45	ns
Rise Time (Fig. 6)	t_r	–	65	ns
Turn-Off Delay (Fig. 7)	t_{d2}	–	60	ns
Fall Time (Fig. 8)	t_f	–	100	ns

($I_D = 2.0$ mAdc, $V_{DS} = 10$ Vdc, ($V_{GS} = 10$ Vdc) (See Figure 9; Times Circuit Determined))

Figure 5.39 2N4351 Motorola *n*-channel enhancement-type MOSFET.

as $V_{GS(Th)}$ and has a range of 1 to 5 V dc, depending on the unit employed. Rather than provide a range of k in Eq. (5.13), a typical level of $I_{D(on)}$ (3 mA in this case) is specified at a particular level of $V_{GS(on)}$ (10 V for the specified I_D level). In other words, when $V_{GS} = 10$ V, $I_D = 3$ mA. The given levels of $V_{GS(Th)}$, $I_{D(on)}$, and $V_{GS(on)}$ permit a determination of k from Eq. (5.14) and a writing of the general equation for the transfer characteristics. The handling requirements of MOSFETs are reviewed in Section 5.9.

EXAMPLE 5.4

Using the data provided on the specification sheet of Fig. 5.39 and an average threshold voltage of $V_{GS(Th)} = 3$ V, determine:
(a) The resulting value of k for the MOSFET.
(b) The transfer characteristics.

Solution

(a) Eq. (5.14): $$k = \frac{I_{D(on)}}{(V_{GS(on)} - V_{GS(Th)})^2}$$
$$= \frac{3 \text{ mA}}{(10 \text{ V} - 3 \text{ V})^2} = \frac{3 \text{ mA}}{(7 \text{ V})^2} = \frac{3 \times 10^{-3}}{49} \text{ A/V}^2$$
$$= \mathbf{0.061 \times 10^{-3} \text{ A/V}^2}$$

(b) Eq. (5.13): $$I_D = k(V_{GS} - V_T)^2$$
$$= 0.061 \times 10^{-3}(V_{GS} - 3 \text{ V})^2$$

For $V_{GS} = 5$ V,

$$I_D = 0.061 \times 10^{-3}(5 \text{ V} - 3 \text{ V})^2 = 0.061 \times 10^{-3}(2)^2$$
$$= 0.061 \times 10^{-3}(4) = 0.244 \text{ mA}$$

For $V_{GS} = 8$, 10, 12, and 14 V, I_D will be 1.525, 3 (as defined), 4.94, and 7.38 mA, respectively. The transfer characteristics are sketched in Fig. 5.40.

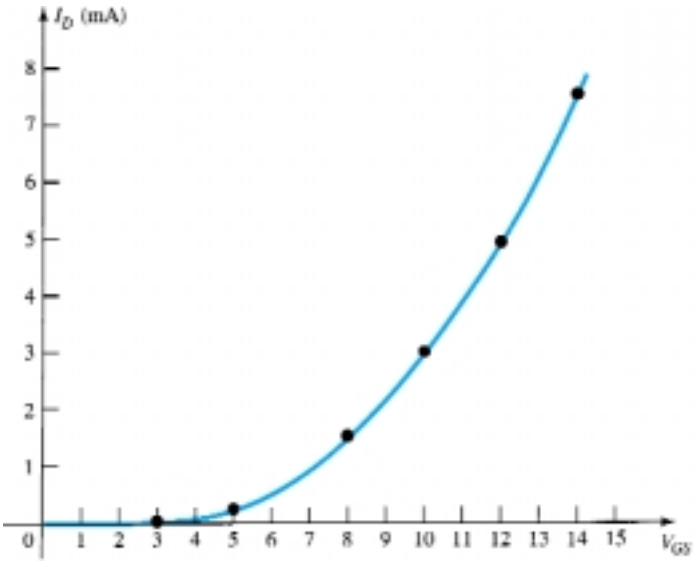


Figure 5.40 Solution to Example 5.4.

5.9 MOSFET HANDLING

The thin SiO_2 layer between the gate and channel of MOSFETs has the positive effect of providing a high-input-impedance characteristic for the device, but because of its extremely thin layer, it introduces a concern for its handling that was not present for the BJT or JFET transistors. There is often sufficient accumulation of static charge (that we pick up from our surroundings) to establish a potential difference across the thin layer that can break down the layer and establish conduction through it. It is therefore imperative that we leave the shorting (or conduction) shipping foil (or ring)

connecting the leads of the device together until the device is to be inserted in the system. The shorting ring prevents the possibility of applying a potential across any two terminals of the device. With the ring the potential difference between any two terminals is maintained at 0 V. At the very least always touch ground to permit discharge of the accumulated static charge before handling the device, and always pick up the transistor by the casing.

There are often transients (sharp changes in voltage or current) in a network when elements are removed or inserted if the power is on. The transient levels can often be more than the device can handle, and therefore the power should always be off when network changes are made.

The maximum gate-to-source voltage is normally provided in the list of maximum ratings of the device. One method of ensuring that this voltage is not exceeded (perhaps by transient effects) for either polarity is to introduce two Zener diodes, as shown in Fig. 5.41. The Zeners are back to back to ensure protection for either polarity. If both are 30-V Zeners and a positive transient of 40 V appears, the lower Zener will "fire" at 30 V and the upper will turn on with a 0-V drop (ideally—for the positive "on" region of a semiconductor diode) across the other diode. The result is a maximum of 30 V for the gate-to-source voltage. One disadvantage introduced by the Zener protection is that the off resistance of a Zener diode is less than the input impedance established by the SiO_2 layer. The result is a reduction in input resistance, but even so it is still high enough for most applications. So many of the discrete devices now have the Zener protection that some of the concerns listed above are not as troublesome. However, it is still best to be somewhat cautious when handling discrete MOSFET devices.

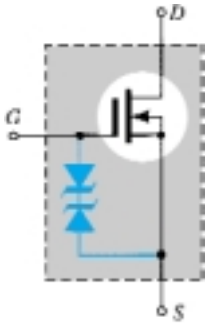


Figure 5.41 Zener-protected MOSFET.

5.10 VMOS

One of the disadvantages of the typical MOSFET is the reduced power-handling levels (typically, less than 1 W) compared to BJT transistors. This shortfall for a device with so many positive characteristics can be softened by changing the construction mode from one of a planar nature such as shown in Fig. 5.23 to one with a vertical structure as shown in Fig. 5.42. All the elements of the planar MOSFET are present in the vertical metal-oxide-silicon FET (VMOS)—the metallic surface connection to the terminals of the device—the SiO_2 layer between the gate and the *p*-type region between the drain and source for the growth of the induced *n*-channel (enhancement-

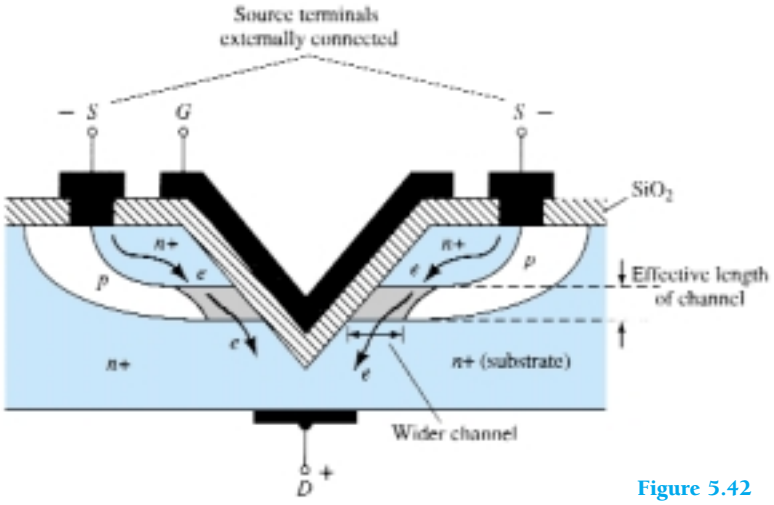


Figure 5.42 VMOS construction.

mode operation). The term *vertical* is due primarily to the fact that the channel is now formed in the vertical direction rather than the horizontal direction for the planar device. However, the channel of Fig. 5.42 also has the appearance of a "V" cut in the semiconductor base, which often stands out as a characteristic for mental memorization of the name of the device. The construction of Fig. 5.42 is somewhat simplistic in nature, leaving out some of the transition levels of doping, but it does permit a description of the most important facets of its operation.

The application of a positive voltage to the drain and a negative voltage to the source with the gate at 0 V or some typical positive "on" level as shown in Fig. 5.42 will result in the induced *n*-channel in the narrow *p*-type region of the device. The length of the channel is now defined by the vertical height of the *p*-region, which can be made significantly less than that of a channel using planar construction. On a horizontal plane the length of the channel is limited to 1 to 2 μm (1 μm = 10^{-6} m). Diffusion layers (such as the *p*-region of Fig. 5.42) can be controlled to small fractions of a micrometer. Since decreasing channel lengths result in reduced resistance levels, the power dissipation level of the device (power lost in the form of heat) at operating current levels will be reduced. In addition, the contact area between the channel and the n^+ region is greatly increased by the vertical mode construction, contributing to a further decrease in the resistance level and an increased area for current between the doping layers. There is also the existence of two conduction paths between drain and source, as shown in Fig. 5.42, to further contribute to a higher current rating. The net result is a device with drain currents that can reach the ampere levels with power levels exceeding 10 W.

In general:

Compared with commercially available planar MOSFETs, VMOS FETs have reduced channel resistance levels and higher current and power ratings.

An additional important characteristic of the vertical construction is:

VMOS FETs have a positive temperature coefficient that will combat the possibility of thermal runaway.

If the temperature of a device should increase due to the surrounding medium or currents of the device, the resistance levels will increase, causing a reduction in drain current rather than an increase as encountered for a conventional device. Negative temperature coefficients result in decreased levels of resistance with increases in temperature that fuel the growing current levels and result in further temperature instability and thermal runaway.

Another positive characteristic of the VMOS configuration is:

The reduced charge storage levels result in faster switching times for VMOS construction compared to those for conventional planar construction.

In fact, VMOS devices typically have switching times less than one-half that encountered for the typical BJT transistor.

5.11 CMOS

A very effective logic circuit can be established by constructing a *p*-channel and an *n*-channel MOSFET on the same substrate as shown in Fig. 5.43. Note the induced *p*-channel on the left and the induced *n*-channel on the right for the *p*- and *n*-channel devices, respectively. The configuration is referred to as a *complementary MOSFET* arrangement (CMOS) that has extensive applications in computer logic design. The relatively high input impedance, fast switching speeds, and lower operating power levels of the CMOS configuration have resulted in a whole new discipline referred to as *CMOS logic design.*

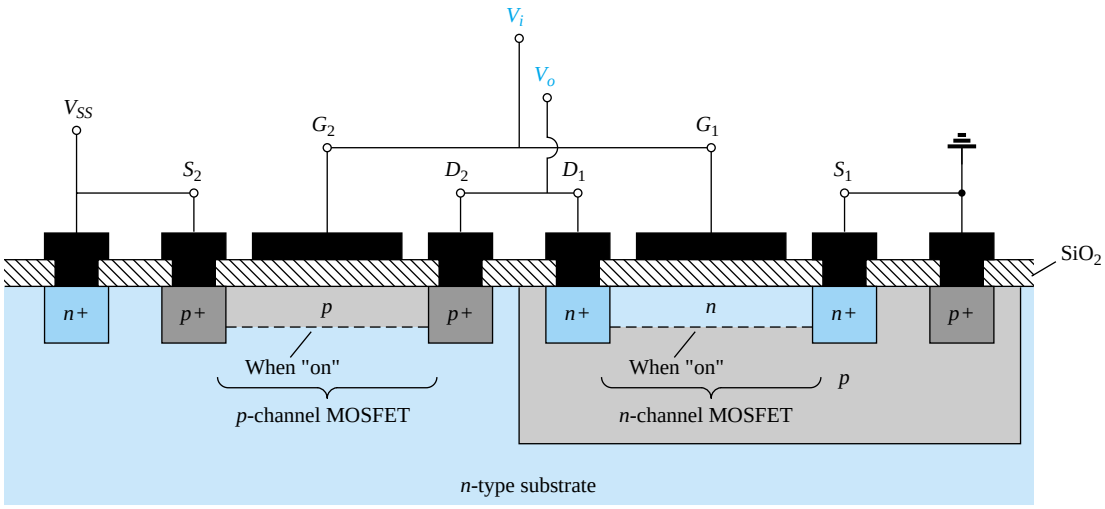


Figure 5.43 CMOS with the connections indicated in Fig. 5.44.

One very effective use of the complementary arrangement is as an inverter, as shown in Fig. 5.44. As introduced for switching transistors, an inverter is a logic element that "inverts" the applied signal. That is, if the logic levels of operation are 0 V (0-state) and 5 V (1-state), an input level of 0 V will result in an output level of 5 V, and vice versa. Note in Fig. 5.44 that both gates are connected to the applied signal and both drain to the output V_o. The source of the *p*-channel MOSFET (Q_2) is connected directly to the applied voltage V_{SS}, while the source of the *n*-channel MOSFET (Q_1) is connected to ground. For the logic levels defined above, the application of 5 V at the input should result in approximately 0 V at the output. With 5 V at V_i (with respect to ground), $V_{GS_1} = V_i$ and Q_1 is "on," resulting in a relatively low resistance between drain and source as shown in Fig. 5.45. Since V_i and V_{SS} are at 5 V, $V_{GS_2} = 0$ V, which is less than the required V_T for the device, resulting in an "off" state. The resulting resistance level between drain and source is quite high for Q_2, as shown in Fig. 5.45. A simple application of the voltage-divider rule will reveal that V_o is very close to 0 V or the 0-state, establishing the desired inversion process. For an applied voltage V_i of 0 V (0-state), $V_{GS_1} = 0$ V and Q_1 will be off with $V_{SS_2} = -5$ V, turning on the *p*-channel MOSFET. The result is that Q_2 will present a small resistance level, Q_1 a high resistance, and $V_o = V_{SS} = 5$ V (the 1-state). Since the drain current that flows for either case is limited by the "off" transistor to the leakage value, the power dissipated by the device in either state is very low. Additional comment on the application of CMOS logic is presented in Chapter 17.

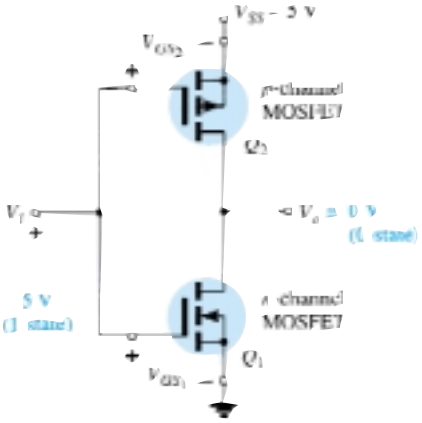


Figure 5.44 CMOS inverter.

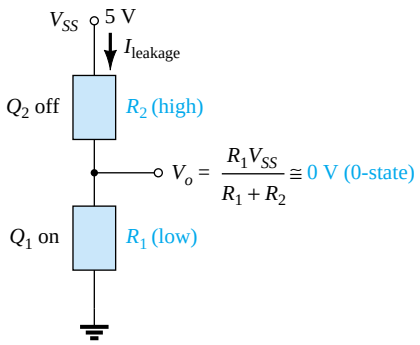
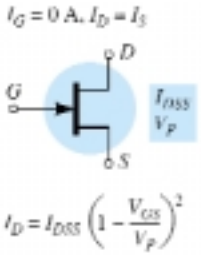
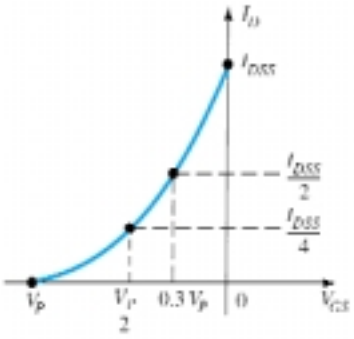
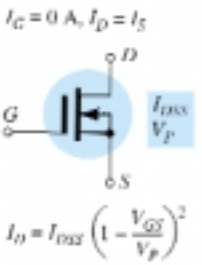
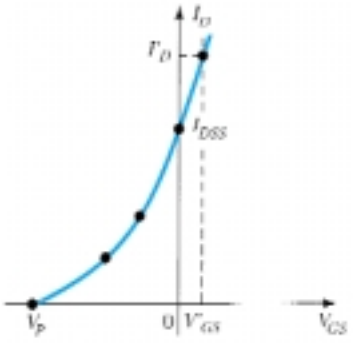
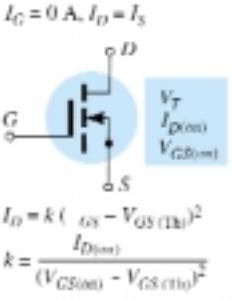
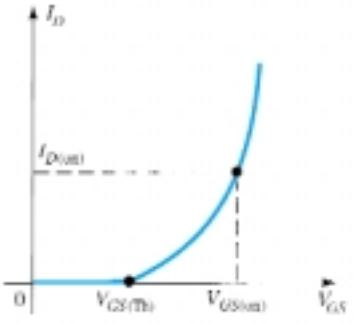


Figure 5.45 Relative resistance levels for $V_i = 5$ V (1-state).

5.12 SUMMARY TABLE

Since the transfer curves and some important characteristics vary from one type of FET to another, Table 5.2 was developed to clearly display the differences from one device to the next. A clear understanding of all the curves and parameters of the table will provide a sufficient background for the dc and ac analyses to follow in Chapters 6 and 8. Take a moment to ensure that each curve is recognizable and its derivation understood, and then establish a basis for comparison of the levels of the important parameters of R_i and C_i for each device.

TABLE 5.2 Field Effect Transistors

Type	*-Symbol- Basic Relationships*	*Transfer Curve*	*Input Resistance and Capacitance*
JFET (*n*-channel)			$R_i > 100$ MΩ C_i: (1 − 10) pF
MOSFET depletion-type (*n*-channel)			$R_i > 10^{10}$ Ω C_i: (1 − 10) pF
MOSFET enhancement-type (*n*-channel)			$R_i > 10^{10}$ Ω C_i: (1 − 10) pF

5.13 PSPICE WINDOWS

The characteristics of an *n*-channel JFET can be found in much the same manner as employed for the bipolar transistor. The series of curves for various levels of V will require a nested sweep under the main sweep for the drain-to-source voltage. The configuration required appears in Fig. 5.46. Note the absence of any resistors since the input impedance is assumed to be infinite, resulting in $I_G = 0$ A. Calling up the device specifications through **Edit-Model-Edit Instance Model (Text)** will result in a display having at the head of the listing a parameter **Beta.** For the junction-field-effect transistor **Beta** is defined by

$$\text{Beta} = \frac{I_{DSS}}{|V_p|^2} \tag{5.15}$$

The parameter **Vto =** -3 defines $V_{GS} = V_P = -3$ V as the pinch-off voltage—something to check when we obtain our characteristics. Choosing the **Setup Analysis** icon (recall that it has the horizontal blue line at the top), the **DC Sweep** is first enabled and then activated to produce the **DC Sweep** dialog box. Select **Voltage Source-Linear,** and insert the **Name:** VDD, the **Start Value** of 0 V, **End Value** of 10 V, and **Increment** of 0.01 V. Then, the **Nested Sweep** is chosen, and **Voltage** and **Linear** are chosen once more. Finally, the **Name:** VGG is entered, the **Start Value** of 0 V is chosen, the **End Value** of -5 V is entered, and the **Increment** is set at -1 V. Then, be sure to **Enable Nested Sweep** before clicking on **OK** and closing. With the **Automatically run Probe after Simulation** enabled, clicking on the analysis icon will result in the **OrCAD-MicroSim Probe** screen. There is no need to call up the **X-Axis Settings** because the horizontal axis has the correct range and the voltage V_{DD} is actually the drain-to-source voltage. By choosing the **Trace** icon, the **Add Traces** dialog box will appear. **ID(J1)** is chosen, followed by **OK.** The result is the set of characteristics appearing in Fig. 5.47. The remaining labels were added using the **ABC** icon.

Note that the pinch-off voltage is -3 V, as expected by the **Vto** parameter. The value of I_{DSS} is very close to 12 mA.

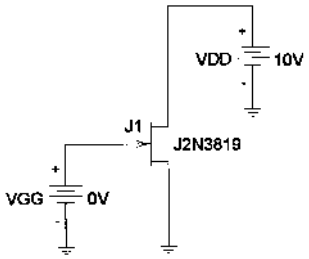


Figure 5.46 Network employed to obtain the characteristics of the *n*-channel J2N3819 JFET.

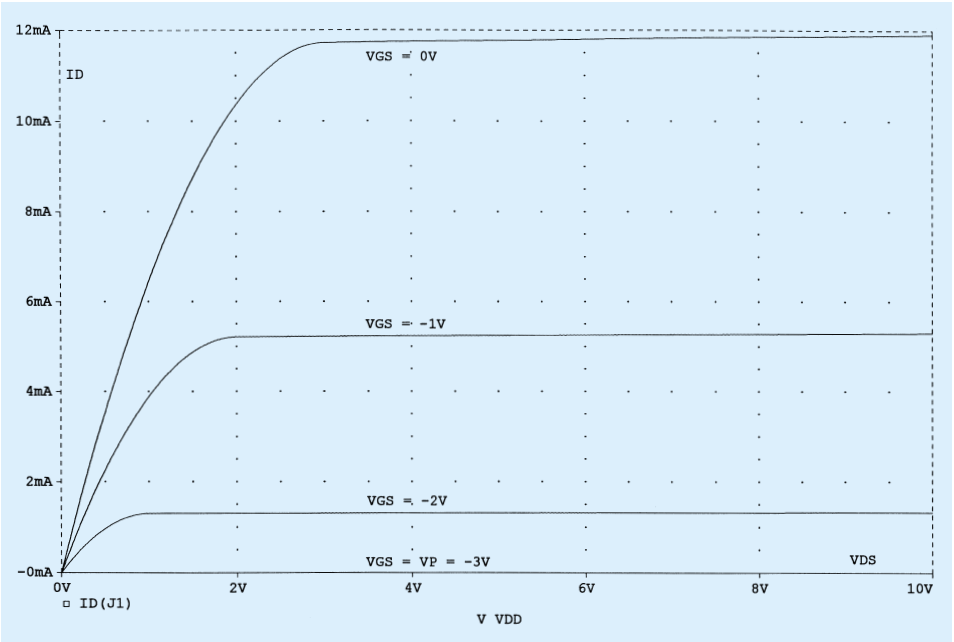


Figure 5.47 Drain characteristics for the *n*-channel J2N3819 JFET of Figure 5.46.

The transfer characteristics can be obtained by returning to the network configuration and choosing the **Analysis-Setup** icon. The **DC Sweep** is again enabled, and the **DC Sweep** is chosen. This time, since the result will only be one curve, a nested operation will not be performed. After choosing **Voltage Source** and **Linear,** the **Name** will be VGG, the **Start Value** -3 V (since we now know that $V_P = -3$ V), the **End Value** 0 V, and the **Increment** 0.01 V to get a good continuous plot. After an **OK** followed by a **Close,** the **Simulation** icon can be chosen. Once the **Probe** screen appears, choose **Plot-X-Axis Settings-Axis Variable** and choose **V(J1:g)** for the gate-to-source voltage. Choose **OK** and we're back to the **X-Axis Settings** dialog box to choose the **User Defined** range of -3 V to 0 V (which already appears because of our sweep settings). Choose **OK** again and the **Trace** ID(J1) can be chosen to result in the transfer characteristics of Fig. 5.48.

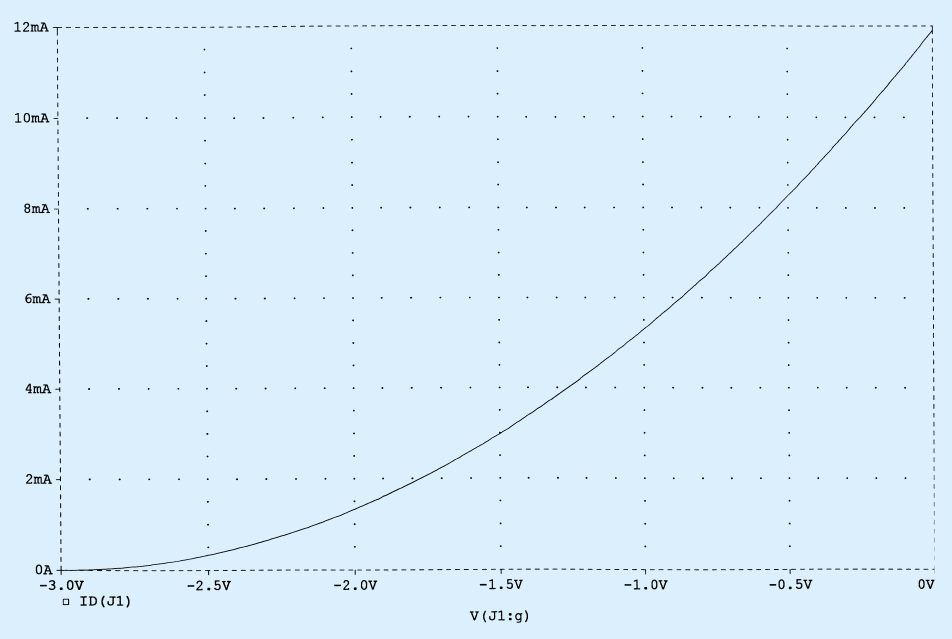


Figure 5.48 Transfer characteristics for the *n*-channel J2N3819 JFET of Figure 5.46.

PROBLEMS

§ 5.2 Construction and Characteristics of JFETs

1. (a) Draw the basic construction of a *p*-channel JFET.
(b) Apply the proper biasing between drain and source and sketch the depletion region for $V_{GS} = 0$ V.

2. Using the characteristics of Fig. 5.10, determine I_D for the following levels of V_{GS} (with $V_{DS} > V_P$).
(a) $V_{GS} = 0$ V.
(b) $V_{GS} = -1$ V.
(c) $V_{GS} = -1.5$ V.
(d) $V_{GS} = -1.8$ V.
(e) $V_{GS} = -4$ V.
(f) $V_{GS} = -6$ V.

3. (a) Determine V_{DS} for $V_{GS} = 0$ V and $I_D = 6$ mA using the characteristics of Fig. 5.10.
(b) Using the results of part (a), calculate the resistance of the JFET for the region $I_D = 0$ to 6 mA for $V_{GS} = 0$ V.
(c) Determine V_{DS} for $V_{GS} = -1$ V and $I_D = 3$ mA.
(d) Using the results of part (c), calculate the resistance of the JFET for the region $I_D = 0$ to 3 mA for $V_{GS} = -1$ V.
(e) Determine V_{DS} for $V_{GS} = -2$ V and $I_D = 1.5$ mA.
(f) Using the results of part (e), calculate the resistance of the JFET for the region $I_D = 0$ to 1.5 mA for $V_{GS} = -2$ V.
(g) Defining the result of part (b) as r_o, determine the resistance for $V_{GS} = -1$ V using Eq. (5.1) and compare with the results of part (d).
(h) Repeat part (g) for $V_{GS} = -2$ V using the same equation, and compare the results with part (f).
(i) Based on the results of parts (g) and (h), does Eq. (5.1) appear to be a valid approximation?

4. Using the characteristics of Fig. 5.10:
(a) Determine the difference in drain current (for $V_{DS} > V_P$) between $V_{GS} = 0$ V and $V_{GS} = -1$ V.
(b) Repeat part (a) between $V_{GS} = -1$ and -2 V.
(c) Repeat part (a) between $V_{GS} = -2$ and -3 V.
(d) Repeat part (a) between $V_{GS} = -3$ and -4 V.
(e) Is there a marked change in the difference in current levels as V_{GS} becomes increasingly negative?
(f) Is the relationship between the change in V_{GS} and the resulting change in I_D linear or nonlinear? Explain.

5. What are the major differences between the collector characteristics of a BJT transistor and the drain characteristics of a JFET transistor? Compare the units of each axis and the controlling variable. How does I_C react to increasing levels of I_B versus changes in I_D to increasingly negative values of V_{GS}? How does the spacing between steps of I_B compare to the spacing between steps of V_{GS}? Compare $V_{C_{sat}}$ to V_P in defining the nonlinear region at low levels of output voltage.

6. (a) Describe in your own words why I_G is effectively zero amperes for a JFET transistor.
(b) Why is the input impedance to a JFET so high?
(c) Why is the terminology *field effect* appropriate for this important three-terminal device?

7. Given $I_{DSS} = 12$ mA and $|V_P| = 6$ V, sketch a probable distribution of characteristic curves for the JFET (similar to Fig. 5.10).

8. In general, comment on the polarity of the various voltages and direction of the currents for an *n*-channel JFET versus a *p*-channel JFET.

§ 5.3 Transfer Characteristics

9. Given the characteristics of Fig. 5.49:
(a) Sketch the transfer characteristics directly from the drain characteristics.
(b) Using Fig. 5.49 to establish the values of I_{DSS} and V_P, sketch the transfer characteristics using Shockley's equation.
(c) Compare the characteristics of parts (a) and (b). Are there any major differences?

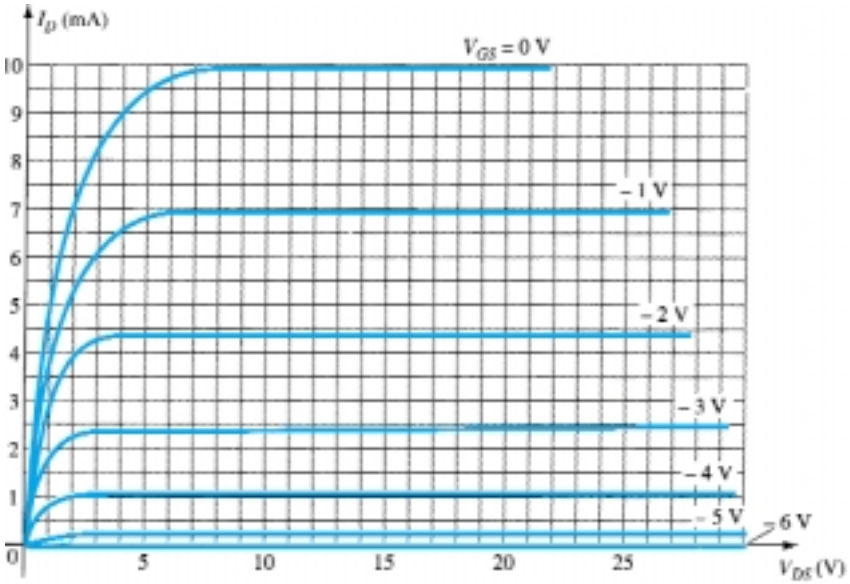


Figure 5.49 Problems 9, 17

10. (a) Given $I_{DSS} = 12$ mA and $V_P = -4$ V, sketch the transfer characteristics for the JFET transistor.
(b) Sketch the drain characteristics for the device of part (a).

11. Given $I_{DSS} = 9$ mA and $V_P = -3.5$ V, determine I_D when:
(a) $V_{GS} = 0$ V.
(b) $V_{GS} = -2$ V.
(c) $V_{GS} = -3.5$ V.
(d) $V_{GS} = -5$ V.

12. Given $I_{DSS} = 16$ mA and $V_P = -5$ V, sketch the transfer characteristics using the data points of Table 5.1. Determine the value of I_D at $V_{GS} = -3$ V from the curve, and compare it to the value determined using Shockley's equation. Repeat the above for $V_{GS} = -1$ V.

13. A *p*-channel JFET has device parameters of $I_{DSS} = 7.5$ mA and $V_P = 4$ V. Sketch the transfer characteristics.

14. Given $I_{DSS} = 6$ mA and $V_P = -4.5$ V:
(a) Determine I_D at $V_{GS} = -2$ and -3.6 V.
(b) Determine V_{GS} at $I_D = 3$ and 5.5 mA.

15. Given a *Q*-point of $I_{D_Q} = 3$ mA and $V_{GS} = -3$ V, determine I_{DSS} if $V_P = -6$ V.

§ 5.4 Specification Sheets (JFETs)

16. Define the region of operation for the 2N5457 JFET of Fig. .5.18 using the range of I_{DSS} and V_P provided. That is, sketch the transfer curve defined by the maximum I_{DSS} and V_P and the transfer curve for the minimum I_{DSS} and V_P. Then, shade in the resulting area between the two curves.

17. Define the region of operation for the JFET of Fig. 5.49 if $V_{DS_{max}} = 25$ V and $P_{D_{max}} = 120$ mW.

§ 5.5 Instrumentation

18. Using the characteristics of Fig. 5.21, determine I_D at $V_{GS} = -0.7$ V and $V_{DS} = 10$ V.

19. Referring to Fig. 5.21, is the locus of pinch-off values defined by the region of $V_{DS} < |V_P| = 3$ V?

20. Determine V_P for the characteristics of Fig. 5.21 using I_{DSS} and I_D at some value of V_{GS}. That is, simply substitute into Shockley's equation and solve for V_P. Compare the result to the assumed value of -3 V from the characteristics.

21. Using $I_{DSS} = 9$ mA and $V_P = -3$ V for the characteristics of Fig. 5.21, calculate I_D at $V_{GS} = -1$ V using Shockley's equation and compare to the level appearing in Fig. 5.21.

22. (a) Calculate the resistance associated with the JFET of Fig. 5.21 for $V_{GS} = 0$ V from $I_D = 0$ to 4 mA.
(b) Repeat part (a) for $V_{GS} = -0.5$ V from $I_D = 0$ to 3 mA.
(c) Assigning the label r_o to the result of part (a) and r_d to that of part (b), use Eq. (5.1) to determine r_d and compare to the result of part (b).

§ 5.7 Depletion-Type MOSFET

23. (a) Sketch the basic construction of a *p*-channel depletion-type MOSFET.
(b) Apply the proper drain-to-source voltage and sketch the flow of electrons for $V_{GS} = 0$ V.

24. In what ways is the construction of a depletion-type MOSFET similar to that of a JFET? In what ways is it different?

25. Explain in your own words why the application of a positive voltage to the gate of an *n*-channel depletion-type MOSFET will result in a drain current exceeding I_{DSS}.

26. Given a depletion-type MOSFET with $I_{DSS} = 6$ mA and $V_P = -3$ V, determine the drain current at $V_{GS} = -1$, 0, 1, and 2 V. Compare the difference in current levels between -1 and 0 V with the difference between 1 and 2 V. In the positive V_{GS} region, does the drain current increase at a significantly higher rate than for negative values? Does the I_D curve become more and more vertical with increasing positive values of V_{GS}? Is there a linear or a nonlinear relationship between I_D and V_{GS}? Explain.

27. Sketch the transfer and drain characteristics of an *n*-channel depletion-type MOSFET with $I_{DSS} = 12$ mA and $V_P = -8$ V for a range of $V_{GS} = -V_P$ to $V_{GS} = 1$ V.

28. Given $I_D = 14$ mA and $V_{GS} = 1$ V, determine V_P if $I_{DSS} = 9.5$ mA for a depletion-type MOSFET.

29. Given $I_D = 4$ mA at $V_{GS} = -2$ V, determine I_{DSS} if $V_P = -5$ V.

30. Using an average value of 2.9 mA for the I_{DSS} of the 2N3797 MOSFET of Fig. 5.30, determine the level of V_{GS} that will result in a maximum drain current of 20 mA if $V_P = -5$ V.

31. If the drain current for the 2N3797 MOSFET of Fig. 5.30 is 8 mA, what is the maximum permissible value of V_{DS} utilizing the maximum power rating?

§ 5.8 Enhancement-Type MOSFET

32. (a) What is the significant difference between the construction of an enhancement-type MOSFET and a depletion-type MOSFET?
(b) Sketch a *p*-channel enhancement-type MOSFET with the proper biasing applied ($V_{DS} > 0$ V, $V_{GS} > V_T$) and indicate the channel, the direction of electron flow, and the resulting depletion region.
(c) In your own words, briefly describe the basic operation of an enhancement-type MOSFET.

33. (a) Sketch the transfer and drain characteristics of an *n*-channel enhancement-type MOSFET if $V_T = 3.5$ V and $k = 0.4 \times 10^{-3}$ A/V^2.
(b) Repeat part (a) for the transfer characteristics if V_T is maintained at 3.5 V but k is increased by 100% to 0.8×10^{-3} A/V^2.

34. (a) Given $V_{GS(Th)} = 4$ V and $I_{D(on)} = 4$ mA at $V_{GS(on)} = 6$ V, determine k and write the general expression for I_D in the format of Eq. (5.13).
(b) Sketch the transfer characteristics for the device of part (a).
(c) Determine I_D for the device of part (a) at $V_{GS} = 2$, 5, and 10 V.

35. Given the transfer characteristics of Fig. 5.50, determine V_T and k and write the general equation for I_D.

36. Given $k = 0.4 \times 10^{-3}$ A/V^2 and $I_{D(on)} = 3$ mA with $V_{GS(on)} = 4$ V, determine V_T.

37. The maximum drain current for the 2N4351 *n*-channel enhancement-type MOSFET is 30 mA. Determine V_{GS} at this current level if $k = 0.06 \times 10^{-3}$ A/V^2 and V_T is the maximum value.

I_{DDS}/V_P

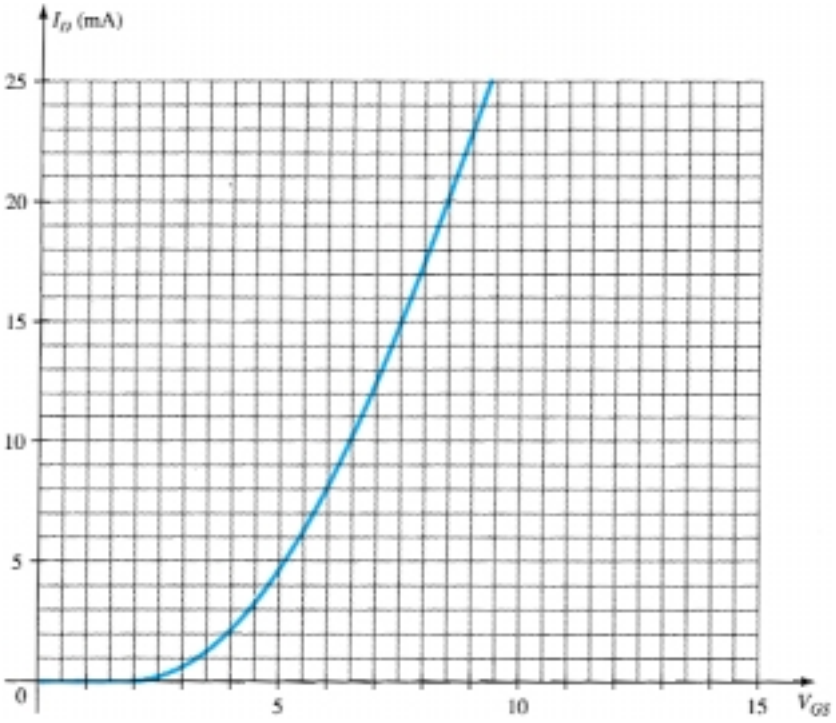


Figure 5.50 Problem 35

38. Does the current of an enhancement-type MOSFET increase at about the same rate as a depletion-type MOSFET for the conduction region? Carefully review the general format of the equations, and if your mathematics background includes differential calculus, calculate dI_D/dV_{GS} and compare its magnitude.

39. Sketch the transfer characteristics of a *p*-channel enhancement-type MOSFET if $V_T = -5$ V and $k = 0.45 \times 10^{-3}$ A/V^2.

40. Sketch the curve of $I_D = 0.5 \times 10^{-3}(V_{GS}^2)$ and $I_D = 0.5 \times 10^{-3}(V_{GS} - 4)^2$ for V_{GS} from 0 to 10 V. Does $V_T = 4$ V have a significant impact on the level of I_D for this region?

§ 5.10 VMOS

41. (a) Describe in your own words why the VMOS FET can withstand a higher current and power rating than the standard construction technique.
(b) Why do VMOS FETs have reduced channel resistance levels?
(c) Why is a positive temperature coefficient desirable?

§ 5.11 CMOS

* **42.** (a) Describe in your own words the operation of the network of Fig. 5.44 with $V_i = 0$ V.
(b) If the "on" MOSFET of Fig. 5.44 (with $V_i = 0$ V) has a drain current of 4 mA with $V_{DS} = 0.1$ V, what is the approximate resistance level of the device? If $I_D = 0.5$ μA for the "off" transistor, what is the approximate resistance of the device? Do the resulting resistance levels suggest that the desired output voltage level will result?

43. Research CMOS logic at your local or college library, and describe the range of applications and basic advantages of the approach.

*Please Note: Asterisks indicate more difficult problems.